# Mean Reversion and Unit Root Properties of Diffusion Models*

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**Abstract**

This paper analyzes the mean reversion and unit root properties of general diffusion models and their discrete samples. In particular, we find that the Dickey-Fuller unit root test applied to discrete samples from a diffusion model becomes a test of no mean reversion rather than a unit root, or more generally, nonstationarity in the underlying diffusion. The unit root test has a well defined limit distribution if and only if the underlying diffusion has no mean reversion, and diverges to minus infinity in probability if and only if the underlying diffusion has mean reversion. It is shown, on the other hand, that diffusions are mean-reverting as long as their drift terms play the dominant role, and therefore, nonstationary diffusions may well have mean reversion.

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# 1 Introduction

The unit root test was originally developed to test for the unity of the largest root in an autoregressive time series model. However, the existence of a unit root entails some other important consequences as well on more general stochastic characteristics of the underlying time series. In particular, it implies that, every shock having a permanent effect, the effects of shocks are accumulated and build up a stochastic trend, which eventually makes any time series with a unit root nonstationary. On the other hand, an autoregressive process becomes stationary if it has the largest root less than unity in modulus, since every shock has only a transitory effect and is offset by a future shock with the opposite sign. Accordingly, the unit root test may also be regarded generally as a test of the nonstationarity or the absence of mean reversion in the underlying time series. In fact, the unit root test has commonly been used to test for the nonstationarity or the absence of mean reversion in a much broader class of time series models than autoregressive models.

Naturally, the unit root test has also been routinely applied to samples that are thought to be collected discretely at relatively high frequencies from diffusion type continuous time processes. As is well known, the laws of motions for economic and financial time series such as stock prices, exchange rates and interest rates, among others, are typically modeled as diffusions. However, the asymptotics of the unit root test applied to discrete samples from general diffusion models have long been unknown, except for the simple case that the underlying diffusion models are driven by Brownian motion and stationary Ornstein-Uhlenbeck processes respectively for the null and alternative hypotheses, for which the reader is referred to Shiller and Perron (1985), Perron (1989, 1991) and Chambers (2004, 2008), among others. In particular, it has been completely unknown whether the unit root test has any discriminatory power in distinguishing general stationary and nonstationary, or mean-reverting and non-mean-reverting, diffusion models. Moreover, the effects of time span and sampling frequency on the power of the unit root test have never been theoretically analyzed for general stationary diffusion models.

In this paper, we investigate the longrun behaviors of general diffusion models including their mean reversion and unit root properties. Our investigation is extensive and complete. There were two major difficulties in establishing our asymptotic theory. First, our theoretical development needs some basic asymptotics for general stationary and nonstationary diffusions. In particular, we need to consider the asymptotics for stationary diffusions having no proper moments and the nonstationary diffusion asymptotics, and they are only recently developed by Kim and Park (2013) up to the generality that is required here. Second, a diffusion model is completely specified by two functions, drift and diffusion functions,

which represent respectively the instantaneous rate of change in conditional mean and the instantaneous conditional volatility. Particularly, they determine all charactersitics of a diffusion model including its asymptotics or longrun behaviors. However, it is not easy to see how the asymptotic properties of a diffusion model are related precisely to the drift and diffusion functions representing its instantaneous characteristics.

A diffusion is either recurrent or transient. A recurrent diffusion visits every point on its domain in finite time no matter where it starts. If there is a nonempty set of points in its domain that it cannot reach in finite time with a positive probability, a diffusion is called transient. In the paper, we mainly consider recurrent diffusions, since they are practically more relevant and yield well defined asymptotics. Diffusions become transient if they have trends exploding to their boundaries. Clearly, if transient diffusions have trends in simple form, we may render them recurrent by detrending. A recurrent diffusion is either positive recurrent or null recurrent. For a positive recurrent diffusion, the time it takes to hit any given level has finite expectation. On the other hand, it may take too long for a diffusion to hit a certain level and the hitting time may have infinite expectation, in which case it becomes null recurrent. It is well known that positive recurrent diffusions are stationary, whereas null recurrent diffusions are nonstationary.

In the paper, we show that the Dickey-Fuller test cannot be used to test for nonstationarity of a diffusion model. The test applied to discrete samples from a diffusion model may diverge to minus infinity in probability for nonstationary diffusions. Indeed, it tests for the absence of mean-reverting behavior in the underlying diffusion. The unit root test has a well defined limit distribution if and only if the underlying diffusion does not have mean reversion, and it diverges to minus infinity in probability if and only if the underlying diffusion has mean reversion. This is true for all recurrent diffusions. On the other hand, all stationary diffusions have mean reversion regardless of the existence of mean, and therefore, the test diverges to minus infinity in probability for all stationary diffusions. However, nonstationary diffusions may also have mean reversion if they have drift terms dominating their diffusion terms. Consequently, the test does not generally discriminate stationary and nonstationary diffusions. Rather, it effectively discriminates mean-reverting and non-mean-reverting diffusions.

Of course, we may use the Dickey-Fuller test to fully discriminate the stationary and nonstationary diffusions, if we only consider nonstationary and non-mean-reverting diffusions under the null hypothesis. As long as the class of diffusion models are restricted as such under the null hypothesis, the test becomes consistent against all stationary diffusion models. Nevertheless, the limit distribution of the test is heavily model-dependent and depends on the underlying diffusion model in a complicated manner. Therefore, the usual

Dickey-Fuller critical values cannot be used even in such a restrictive setup. The limit distribution of the test is generally represented as a functional of a skew Bessel process, and reduces to the Dickey-Fuller distribution only if the underlying diffusion becomes a Brownian motion in the limit. The test relying on the Dickey-Fuller distribution has a correct asymptotic size only in this case. Consequently, the Dickey-Fuller test is valid only if we assume that the underlying diffusion is asymptotically Brownian motion.

The rest of the paper is organized as follows. Section 2 presents the background and preliminaries that are necessary to understand the subsequent development of asymptotic theory developed in the paper. The diffusion model and various notions to investigate its longrun behaviors, and regularly and rapidly varying functions with their useful properties are introduced. In Section 3, we establish the relationship between the longrun behaviors, such as recurrence and stationarity, of a diffusion model with its drift and diffusion functions, which are instantaneous parameters representing respectively the rate of change in conditional mean and volatility. Section 4 explores the mean-reverting behaviors of diffusion models, and establishes the necessary and sufficient condition for the mean reversion of diffusion model. We consider the Dickey-Fuller unit root test and develop its asymptotics in Section 5, and show that the unit root test is indeed a test for the absence of mean reversion. Section 6 concludes the paper, and all mathematical proofs are in Appendix.

Finally, a word on notation. Since our asymptotics are extensive and complicated, we need to make some notational conventions to facilitate our exposition. The notation "$\sim$" signifies the asymptotic equivalence. In particular, $f(x) \sim g(x)$ at $\underline{x}$ or $\overline{x}$ implies that $f(x)/g(x) \to 1$ as $x \to \underline{x}$ or $x \to \overline{x}$, respectively. On the other hand, "$P_T \sim_p Q_T$" and "$P_T \prec_p Q_T$" just imply that $P_T/Q_T \to_p 1$ and $P_T = o_p(Q_T)$, respectively, as $T \to \infty$. Moreover, we follow the notational convention in the markov process literature to use the same notation for both a measure and its density with respect to Lebesgue measure. As an example, for a given measure or a density $m$ on $\mathcal{D} \subset \mathbb{R}$, we write $m(\mathcal{D})$ and $m(f)$ interchangeably with $\int_{\mathcal{D}} m(x)dx$ and $\int_{\mathcal{D}} m(x)f(x)dx$ respectively whenever they are well defined. Furthermore, as usual, $1\{\cdot\}$ is defined as the indicator function. The identity function on $\mathbb{R}$ is denoted by $\iota$, i.e., $\iota(x) = x$, and we write the $\kappa$-th order power function as $\iota^\kappa$ so that $\iota^\kappa(x) = x^\kappa$.

# 2 Background and Preliminaries

## 2.1 Diffusion Model

We consider the diffusion process $X$ given by the time-homogeneous stochastic differential equation

$$dX_t = \mu(X_t)dt + \sigma(X_t)dW_t, \tag{1}$$

where $\mu$ and $\sigma$ are, respectively, the drift and diffusion functions, and $W$ is the standard Brownian motion. We denote by $\mathcal{D} = (\underline{x}, \overline{x})$ the domain of the diffusion process $X$, where we set $\underline{x} = -\infty$ or 0 with $\overline{x} = \infty$. This causes no loss in generality, since we may simply consider $X - \underline{x}$ or $-X$ to allow for a more general case. In what follows, we denote by $x_B = \underline{x}$ or $\overline{x}$ the boundary of $\mathcal{D}$. Throughout the paper, we assume

**Assumption 2.1.** *We assume that (a) $\sigma^2(x) > 0$ for all $x \in \mathcal{D}$, and (b) $\mu(x)/\sigma^2(x)$ and $1/\sigma^2(x)$ are locally integrable at every $x \in \mathcal{D}$.*

Assumption 2.1 provides a simple sufficient set of conditions to ensure that a weak solution to the stochastic differential equation (1) exists uniquely up to an explosion time. See, e.g., Theorem 5.5.15 in Karatzas and Shreve (1991).

The scale function of the diffusion process $X$ in (1) is defined as

$$s(x) = \int_w^x \exp\left(-\int_w^y \frac{2\mu(z)}{\sigma^2(z)}dz\right) dy, \tag{2}$$

where the lower limits of the integrals can be arbitrarily chosen to be any point $w \in \mathcal{D}$. Defined as such, the scale function $s$ is uniquely identified up to any increasing affine transformation, i.e., if $s$ is a scale function, then so is $as + b$ for any constants $a > 0$ and $-\infty < b < \infty$. We also define the speed density

$$m(x) = \frac{1}{(\sigma^2 s')(x)} \tag{3}$$

on $\mathcal{D}$, where $s'$ is the derivative of $s$, often called the scale density, which is assumed to exist. The speed measure is defined to be the measure on $\mathcal{D}$ given by the speed density with respect to the Lebesgue measure. Note, under Assumption 2.1, that both the scale function and speed density are well defined, and that the scale function is strictly increasing, on $\mathcal{D}$.

Our asymptotic theory depends crucially on the recurrence property of the diffusion process $X$. To define the recurrence property, we let $\tau_y$ be the hitting time of a point $y$ in $\mathcal{D}$ that is given by $\tau_y = \inf\{t \geq 0 | X_t = y\}$. We say that the diffusion $X$ is recurrent if $\mathbb{P}\{\tau_y < \infty | X_0 = x\} = 1$ for all $x, y \in \mathcal{D}$. The recurrent diffusion $X$ is said to be positive recurrent

if $\mathbb{E}[\tau_y < \infty | X_0 = x] < \infty$ for all $x, y \in \mathcal{D}$, and null recurrent if $\mathbb{E}[\tau_y < \infty | X_0 = x] = \infty$ for all $x, y \in \mathcal{D}$. Under some regularity conditions on $\mu$ and $\sigma$, the diffusion $X$ is recurrent if and only if the scale function $s$ in (2) is unbounded at both boundaries $\underline{x}$ and $\overline{x}$, i.e.,

$$s(\underline{x}) = -\infty \quad \text{and} \quad s(\overline{x}) = \infty.$$

Furthermore, the recurrent diffusion $X$ becomes positive recurrent or null recurrent, depending upon

$$m(\mathcal{D}) < \infty \quad \text{or} \quad m(\mathcal{D}) = \infty,$$

where $m$ is the speed measure defined in (3). A diffusion which is not recurrent is said to be transient.

Positive recurrent diffusions are stationary. More precisely, they have time invariant distributions, and if they are started from the time invariant distributions they become stationary. The time invariant density of the positive recurrent diffusion $X$ is given by

$$\pi(x) = \frac{m(x)}{m(\mathcal{D})}.$$

Null recurrent and transient diffusions are nonstationary. They do not have time invariant distributions, and their marginal distributions change over time. Out of these two different types of nonstationary processes, we mainly consider null recurrent diffusions in the paper. Brownian motion is the prime example of null recurrent diffusions. Typically, transient processes have upward or downward trends, in which case we may eliminate their trends using appropriate detrending methods so that they behave like recurrent processes. Like unit root processes in discrete time, null recurrent processes have stochastic trends and the standard law of large numbers and central limit theory in continuous time are not applicable. See, e.g., Jeong and Park (2011) and Kim and Park (2013) for more details on the statistical properties of null recurrent diffusions.

## 2.2 Regularly and Rapidly Varying Functions

In the paper, we use extensively the concepts of regularly and rapidly varying functions. For a function $f : (0, \infty) \to \mathbb{R}$, we say that it is *regularly varying* at infinity with index $\kappa$, and write $f \in RV_\kappa$, if for all $x > 0$ $\lim_{\lambda \to \infty} f(\lambda x)/f(\lambda) = x^\kappa$ with some $\kappa \in (-\infty, \infty)$. In particular, if $\kappa = 0$ and $f \in RV_0$, then $f$ is said to be *slowly varying* at infinity. Moreover,

we say that $f$ is *rapidly varying* at infinity with index $\infty$ or $-\infty$, if

$$\liminf_{\lambda\to\infty}\inf_{x\geq 1}\frac{f(\lambda x)}{x^{\kappa}f(\lambda)}\geq 1 \quad \text{or} \quad \limsup_{\lambda\to\infty}\sup_{x\geq 1}\frac{f(\lambda x)}{x^{\kappa}f(\lambda)}\leq 1,$$

in which case we write $f\in RV_{\infty}$ or $f\in RV_{-\infty}$, respectively. See Bingham et al. (1993) for more discussions on regularly and rapidly varying functions, as well as their alternative concepts and definitions. In our subsequent discussions, we will simply write $f\in RV_{\kappa}$ with $\kappa\in[-\infty,\infty]$, if $f$ is regularly or rapidly varying.

It is necessary to define the notions of regular and rapid variations at $-\infty$ and 0, as well as at $\infty$, in our paper. To extend the notions of regular and rapid variations, we just follow the usual convention. For the notions of regular and rapid variations at $-\infty$ for $f$ on $(-\infty,\infty)$, we define $g$ from the negative part of $f$ as $g(x)=f(-x)$, and define $f$ to be regularly and rapidly varying at $-\infty$ with index $\kappa$ if and only if $g$ is regularly and rapidly varying at $\infty$ with index $\kappa$. On the other hand, for the notions of regular and rapid variations at 0 for $f$ on $(0,\infty)$, we define $g$ from $f$ as $g(x)=f(1/x)$, and define $f$ to be regularly and rapidly varying at 0 with index $-\kappa$ if and only if $g$ is regularly and rapidly varying at $\infty$ with index $\kappa$.

Throughout the paper, we make the following convention. For function $f:\mathcal{D}\to\mathbb{R}$ with $\mathcal{D}=(\underline{x},\overline{x})$, we write $f\in RV_{\kappa}$ with two dimensional index $\kappa$ defined as $\kappa=(\underline{\kappa},\overline{\kappa})$, where $\underline{\kappa}\in[-\infty,\infty]$ and $\overline{\kappa}\in[-\infty,\infty]$ are indexes of regular and rapid variations of $f$ at $\underline{x}$ and $\overline{x}$. We let $x_B$ be either $\underline{x}$ or $\overline{x}$ and, once $x_B$ is designated as $\underline{x}$ or $\overline{x}$, we use $\kappa$ to denote the index of regular or rapid variation at the corresponding boundary, i.e., $\underline{\kappa}$ or $\overline{\kappa}$. For instance, we write $\kappa=1$ at $x_B=0$, in place of $\underline{\kappa}=1$ at $\underline{x}=0$, should there be no confusion. Moreover, equalities and inequalities on $\kappa$ apply to both $\underline{\kappa}$ and $\overline{\kappa}$, and therefore, $\kappa=1$ implies $\underline{\kappa}=\overline{\kappa}=1$, and $\kappa>1$ implies $\underline{\kappa}>1$ and $\overline{\kappa}>1$, respectively.

To effectively deal with the asymptotics of $f\in RV_{\kappa}$, $\kappa=(\underline{\kappa},\overline{\kappa})$, on $\mathcal{D}=(-\infty,\infty)$, we make a notational convention and let

$$f(\lambda)=|f(\lambda)|+|f(-\lambda)| \tag{4}$$

for any numerical sequence $\lambda\to\infty$, so that we have

$$\frac{f(\lambda x)}{f(\lambda)}\to\overline{f}(x)$$

as $\lambda \to \infty$, where $\overline{f}$, called the *limit homogeneous function*, is given by

$$\overline{f}(x) = a|x|^{\kappa}1\{x > 0\} + b|x|^{\kappa}1\{x < 0\}$$

for some constant $a, b \in \mathbb{R}$ such that $|a| + |b| = 1$. The convention (4) greatly simplifies our exposition and will be made throughout the paper without further reference. Of course, our convention becomes unnecessary and redundant if we consider nonnegative $f$ defined on $(0, \infty)$.

There are many useful properties of regularly and rapidly varying functions. At any $x_B$, we have (i) $f_1 \in RV_{\kappa_1}$ and $f_2 \in RV_{\kappa_2}$ implies $f_1 f_2 \in RV_{\kappa_1+\kappa_2}$ as long as $\kappa_1 + \kappa_2$ is well defined, (ii) $f \in RV_{\kappa}$ implies $1/f \in RV_{-\kappa}$, if $f > 0$, and (iii) $f \in RV_{\kappa}$ implies $f^{-1} \in RV_{1/\kappa}$, where $f^{-1}$ is the generalized inverse of $f$. Also, it follows from the Karamata's representation theorem that $f : \mathcal{D} \to \mathbb{R}$ is $RV_{\kappa}$ at $x_B$ if it may be written in the form

$$f(x) = a(x) \exp\left(\int^{x} \frac{b(y)}{y} dy\right)$$

for $a(x) \to c$ and $b(x) \to \kappa$ as $x \to x_B$. Furthermore, any regularly varying function allows for a simple asymptotic expansion of integrals, which is known as the Karamata's theorem, and we have for any $f \in RV_{\kappa}$

$$\int^{x} f(y)dy \sim \begin{cases} xf(x)/(\kappa+1), & \text{if } \kappa > -1 \\ \ell(x), & \text{if } \kappa = -1 \\ c, & \text{if } \kappa < -1 \end{cases}$$

at $x_B = \pm\infty$, and

$$\int^{x} f(y)dy \sim \begin{cases} xf(x)/(\kappa+1), & \text{if } \kappa < -1 \\ \ell(x), & \text{if } \kappa = -1 \\ c, & \text{if } \kappa > -1 \end{cases}$$

at $x_B = 0$, where for some slowly varying function $\ell$ and constant $c$. The reader is referred to Bingham et al. (1993) for more details of the regularly and rapidly varying functions.

## 3 Diffusion Asymptotics

A diffusion is defined by its drift and diffusion functions. Though drift and diffusion functions are infinitesmal parameters that primarily describe instantaneous dynamics, they completely specify a diffusion, including its longrun behavior. The asymptotics of a diffusion

such as recurrence and stationarity, however, are not to be readily obtained from its drift and diffusion functions, since it is not clearly seen how drift and diffusion functions determine the longrun behavior of a diffusion. In this section, we develop a new framework we may conveniently use to find the asymptotics of a diffusion.

To analyze diffusion asymptotics, it is very useful to introduce a function $v : \mathcal{D} \to \mathbb{R}$ defined as

$$v(x) = -\frac{2x\mu(x)}{\sigma^2(x)},$$

which we call the *scale index function* in the paper. Note that the scale density $s'$, which is well defined by Assumption 2.1, can be written in the form

$$s'(x) = \exp\left(\int^x \frac{v(y)}{y} dy\right). \tag{5}$$

We assume

**Assumption 3.1.** *(a) $v$ has limit, and (b) $\sigma^2$ is regularly varying.*

Let $p = (\underline{p}, \overline{p})$, $\underline{p}, \overline{p} \in [-\infty, \infty]$, be the limit of scale index function $\nu$ as $x \to x_B = (\underline{x}, \overline{x})$, which plays an important role in diffusion asymptotics. Under Assumption 3.1 (a), it follows directly from (5) that

$$s' \in RV_p, \tag{6}$$

due to the Karamata's representation theorem. Given (6), Assumption 3.1 (b) implies that $m$ is also regularly or rapidly varying, due to well known properties of regularly and rapidly varying functions, and we write

$$m \in RV_q, \tag{7}$$

where $q = (\underline{q}, \overline{q})$, $\underline{q}, \overline{q} \in [-\infty, \infty]$. Under Assumption 3.1, the recurrence and stationarity properties of a diffusion may be readily obtained from two indexes $p$ and $q$ introduced in (6) and (7).

**Assumption 3.2.** $p \neq -1$.

In what follows, we call a diffusion *regular* if it satisfies Assumption 3.2. The recurrence of an irregular diffusion is determined by the slowly varying component of $s'$, and becomes very complicated to analyze. Moreover, an irregular diffusion has a slowly varying scale function with a rapidly varying inverse. In the paper, we only analyze diffusion with a scale function that has a regularly varying inverse. As discussed in Kim and Park (2013), the asymptotics of a diffusion having a scale function whose inverse is rapidly varying become nonstandard and generally path-dependent.

For a regular diffusion, it follows immediately from the Karamata's theorem that we have $s(\pm\infty) = \pm\infty$ if and only if $p > -1$, and $s(0) = -\infty$ if and only if $p < -1$. Therefore, a regular diffusion on $\mathcal{D} = (-\infty, \infty)$ is recurrent if and only if $p > -1$. Likewise, a regular diffusion on $\mathcal{D} = (0, \infty)$ is recurrent if and only if $\underline{p} < -1$ and $\overline{p} > -1$. The stationarity of a recurrent diffusion is given by the integrability of $m$. Note that $m$ is integrable if $q < -1$ at boundaries $\pm\infty$, and if $q > -1$ at boundary 0. Consequently, a recurrent diffusion on $\mathcal{D} = (-\infty, \infty)$ is positive recurrent if $q < -1$. By the same token, a recurrent diffusion on $\mathcal{D} = (0, \infty)$ is positive recurrent if $\underline{q} > -1$ and $\overline{q} < -1$. If $q = -1$, however, the stationarity of a recurrent diffusion is determined by the slowly varying component of $m$.

It is straightforward to determine the recurrence and stationarity of any given diffusion using our approach.

**Example 3.1.** The nonlinear drift diffusion (NLD) on $\mathcal{D} = (0, \infty)$ is introduced by Aït-Sahalia (1996) as a model for spot interest rates. The diffusion is specified as

$$dX_t = \left(a_0 + a_1 X_t + a_2 X_t^2 + a_{-1} X_t^{-1}\right) dt + \sqrt{b_0 + b_1 X_t + b_2 X_t^2} dW_t. \tag{8}$$

Aït-Sahalia (1996) derives the conditions for the process to be positive recurrent process, by explicitly computing boundary values of $s$ and directly examining integrability of $m$. The actual derivation is not simple and rather involved. However, his conditions can be readily obtained if our approach is taken, since it only relies on the indexes $(p, q)$ of $s' \in RV_p$ and $m \in RV_q$ at both boundaries. Assuming leading coefficients $a_{-1}, b_0$ and $a_2, b_2$ respectively for boundaries $x_B = 0$ and $\infty$ are nonzero in (8), we may easily find that

$$\underline{p} = -\frac{2a_{-1}}{b_0} \quad \text{and} \quad \overline{p} = \begin{cases} \infty, & \text{if } a_2/b_2 < 0 \\ -\infty, & \text{if } a_2/b_2 > 0 \end{cases}$$

and $\underline{q} = -\underline{p}$ and $\overline{q} = -(\overline{p} + 2)$. Recall that, in the absence of slowly varying component, the necessary and sufficient condition is $\underline{p} < -1$ and $\overline{p} > -1$ for recurrence, and $\underline{q} > -1$ and $\overline{q} < -1$ for stationarity. Therefore, we may deduce that the NLD process is positive recurrent if and only if $2a_{-1}/b_0 > 1$ and $a_2/b_2 < 0$. Note that the stationarity condition is automatically satisfied if the recurrence condition holds.

**Example 3.2.** We consider the diffusion model on $\mathbb{R}$ given by

$$dX_t = aX_t(1 + X_t^2)^{b-1} dt + \sqrt{(1 + X_t^2)^b} dW_t \tag{9}$$

with parameters $a, b \in \mathbb{R}$, which we call the generalized Höpfner and Kutoyants model. If $b = 0$ or $b = 1$, it reduces to the model considered by Höpfner and Kutoyants (2003) or Chen

et al. (2010), respectively. In particular, if $a = 0$, then the diffusion becomes the driftless diffusion. Clearly, diffusion (9) satisfies Assumption 2.1, and hence, the weak solution exists uniquely up to an explosion time. It can be shown that the scale density and speed measure of the generalized Höpfner and Kutoyants model are simply given by

$$s'(x) = (1 + x^2)^{-a} \quad \text{and} \quad m(x) = (1 + x^2)^{a-b}.$$

For the diffusion, we have $p = -2a$ and $q = 2a - 2b$, and therefore, if we set $a < 1/2$ it becomes recurrent on $\mathbb{R}$. Moreover, the diffusion is stationary if and only if $q < -1$, which holds if and only if $a - b < -1/2$. Note that the recurrence and stationarity conditions become identical if and only if $b = 0$.

Diffusion asymptotics can be obtained directly from the functional forms of $\mu$ and $\sigma^2$ if they are regularly varying. To show this, we write

$$\mu(x) \sim a \operatorname{sgn}(x)|x|^z \ell_1(x) \quad \text{and} \quad \sigma^2(x) \sim b\,|x|^w \ell_2(x) \tag{10}$$

as $x \to x_B$ for some constants $a, b, z$ and $w$, where we assume that $a$ and $b > 0$ are defined in such a way that slowly varying functions $\ell_1$ and $\ell_2$ are both positive near $x_B$ and $\ell(x) = \ell_1(x)/\ell_2(x) \to 1$ as $x \to x_B$ unless it converges to 0 or diverges to $\infty$. We may easily deduce from (10) that

$$v(x) \sim \frac{-2a}{b}\,|x|^{z-w+1}\ell(x) \tag{11}$$

as $x \to x_B$. In our subsequent discussions, we assume $a \neq 0$.

Momentarily, we focus only on the cases of $p = 0$, $p = \infty$ or $p = -\infty$. At $x_B = \pm\infty$, the recurrence condition is satisfied only when $p = 0$ or $p = \infty$, which holds if and only if VI or DI holds, respectively, where

(VI) $z - w + 1 < 0$ (or $z - w + 1 = 0$ and $\ell(x) \to 0$), and
(DI) $z - w + 1 > 0$ (or $z - w + 1 = 0$ and $\ell(x) \to \infty$) and $a < 0$.

On the other hand, at $x_B = 0$, the recurrence condition is satisfied only when $p = -\infty$, which holds if and only if DO holds, where

(DO) $z - w + 1 < 0$ (or $z - w + 1 = 0$ and $\ell(x) \to 0$) and $a > 0$.

Clearly, the underlying diffusion becomes recurrent only when the recurrence condition holds at both boundaries, and otherwise it becomes transient.

Once we have any of recurrence conditions VI, DI or DO, the stationarity condition can also be easily checked. Under VI, the scale density function $s'$ becomes slowly varying. Therefore, unless $w = 1$, the integrability of $m = 1/(s'\sigma^2)$ is completely determined by

the order of $1/\sigma^2$. As a consequence, the stationarity condition holds and fails to hold, if $w > 1$ and $w < 1$, respectively. Note in particular that, under VI, $z > 0$ implies $w > 1$, and therefore, we always have positive recurrence. Clearly, if $w = 1$, then we have $m \in RV_q$ with $q = -1$, and the slowly varying part of $m$ determines the stationarity of the underlying diffusion. Under DI and DO, on the other hand, the stationarity condition is always fulfilled at $x_B = \pm\infty$ and $x_B = 0$ as long as $\sigma^2$ is regularly varying, since $p = \infty$ and $p = -\infty$, respectively. Consequently, if DI or DO holds, we have positive recurrence, and otherwise, we have transience. No null recurrence appears. Again, the underlying diffusion becomes stationary only when the stationarity condition holds at both boundaries, and otherwise it becomes nonstationary.

Now let

(DV) $z - w + 1 = 0$ and $\ell(x) \to 1$

as $x \to x_B$, so that we have $p = -2a/b$ at $x_B$. Under DV, the necessary and sufficient condition for recurrence of any regular diffusion becomes $2a/b < 1$ at $x_B = \pm\infty$, and $2a/b > 1$ at $x_B = 0$. Moreover, we have $m \in RV_q$ with

$$q = \frac{2a}{b} - w = \frac{2a}{b} - (z + 1)$$

at $x_B$. It therefore follows that the stationarity condition holds at $x_B = \pm\infty$ if $2a/b < z$, and at $x_B = 0$ if $2a/b > z$. Consequently, at $x_B = \pm\infty$ we have

| | | |
|---|---|---|
| positive recurrence | | $2a/b < z$ |
| null recurrence | if | $z < 2a/b < 1$ |
| transience | | $2a/b > 1$ |

for $z < 1$, and at $x_B = 0$ we have

| | | |
|---|---|---|
| positive recurrence | | $2a/b > z$ |
| null recurrence | if | $z > 2a/b > 1$ |
| transience | | $2a/b < 1$ |

for $z > 1$. If $z \geq 1$ at $x_B = \pm\infty$ or $z \leq 1$ at $x_B = 0$, any regular diffusion is either positive recurrent or transient. In this case, no regular diffusion may be null recurrent. In particular, any regular diffusion with linear drift and regularly varying volatility is positive recurrent if and only if it is recurrent. In particular, no null recurrent diffusion may have linear drift.

In Figure 1, we characterize the diffusion asymptotics by $(a, b, z, w)$. The top and bottom panels illustrate the diffusion asymptotics respectively at $x_B = \pm\infty$ and $x_B = 0$.

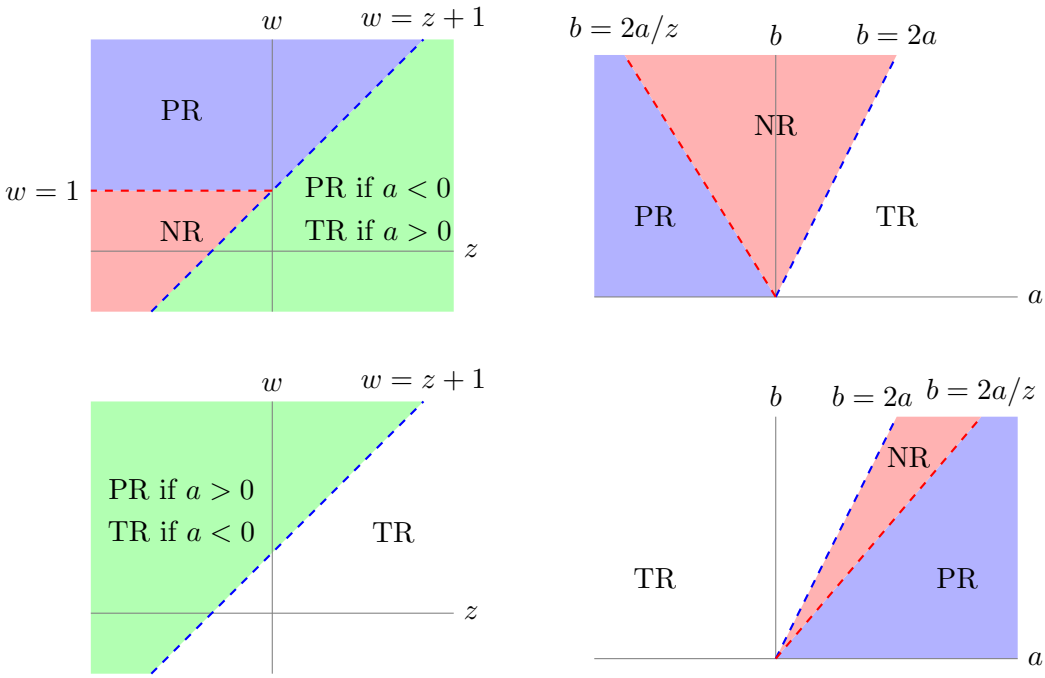



Figure 1: Diffusion asymptotics by $(a, b, z, w)$ at $x_B = \pm\infty$ (top panels) and at $x_B = 0$ (bottom panels).

In the left top panel, the green area represents the condition DI, whereas the blue and red areas represent the condition VI. We have stationarity on the green and blue areas, and nonstationarity on the red area. The right top panel illustrates DV at $x_B = \pm\infty$, and we have stationarity and nonstationarity on the blue and red areas, respectively. Similarly, the green area in the left bottom panel represents DO, and we have stationarity. Moreover, the right bottom panel illustrates DV at $x_B = 0$, and the blue and red areas represent stationarity and nonstationarity, respectively.

## 4 Mean Reversion

In this section, we investigate the mean-reverting behavior of diffusion. The mean-reverting behavior of a diffusion is closely related to the absence of a unit root in the discrete samples collected from it. Indeed, we show later in the next section that a diffusion has mean reversion if and only if it does not have a unit root in its discrete samples. If applied to discrete samples from a diffusion, the unit root test becomes a test for the absence of mean reversion rather than a test for the nonstationarity. Below we introduce a formal definition of mean reversion. Here and elsewhere in the paper, we let $\overline{X} = (\overline{X}_t)$, $\overline{X}_t = t^{-1} \int_0^t X_s ds$,

be the sample mean process of $X$.

**Definition 4.1.** A diffusion $X$ is said to have *mean reversion* if

$$\frac{1}{C_T}\int_0^T (X_t - \overline{X}_T)dX_t \to_d Z$$

as $T \to \infty$ for some normalizing sequence $(C_T)$, where $Z$ is a random variable with support on a subset of $(-\infty, 0)$.

For the mean-reverting diffusion $X$ as defined in Definition 4.1, we say that it has *strong* mean reversion if $Z$ has a point support, and *weak* mean reversion otherwise.

The motivation for our definition of mean reversion is clear. We have

$$M_T = \int_0^T (X_t - \overline{X}_T)dX_t \approx \sum_{i=1}^n (X_{t_{i-1}} - \overline{X}_{t_n})(X_{t_i} - X_{t_{i-1}})$$

for $0 = t_0 < \cdots < t_n = T$ with $\max_{1\le r\le n} |t_i - t_{i-1}| \approx 0$. Therefore, negative $M_T$ implies that $(X_{t_{i-1}} - \overline{X}_{t_n})$ has a negative sample correlation with $(X_{t_i} - X_{t_{i-1}})$, i.e., the deviation from sample mean in the current period is negatively correlated with the increment made in transition to the next period. This occurs if and only if whenever $X$ is observed below its sample mean it has tendency to increase, and vice versa. In describing the mean-reverting behavior of diffusion, we may use the recursive mean and define $(X_t - \overline{X}_t)$ as the deviation from mean, in place of $(X_t - \overline{X}_T)$ relying on the sample mean $\overline{X}_T$ over the entire time span. Though we do not show explicitly in the paper, all our subsequent theories are also applicable for this alternative definition of mean reversion only with some obvious minor changes.

To analyze the mean reversion property, it is necessary to introduce the asymptotics for additive functionals of continuous time process $\int_0^T f(X_t)dt$ as $T \to \infty$. The asymptotics are crucially dependent upon the integrability of $f$ in $m$, or simply, the $m$-integrability. In case that $f$ is $m$-integrable, the required asymptotics are already well known, as we may find in, e.g., Rogers and Williams (2000) and Höpfner and Löcherbach (2003) respectively for positive recurrent and null recurrent diffusions. Recently, Kim and Park (2013) have developed in a unified framework the complete asymptotic theory of additive functions of a diffusion for $m$-non integrable $f$, as well as $m$-integrable $f$, for both positive and null recurrent diffusions. Here we use their asymptotics.

With no loss of generality, we may write $\int_0^T f(X_t)dt = \int_0^T f_s(X_t^s)dt$, where $f_s = f \circ s^{-1}$ and $X^s = s(X)$ is the scale transformation of $X$, which may be defined as $dX_t^s =$

$m_s^{-1/2}(X_t^s)dW_t$ with speed measure $m_s$ given by

$$m_s = \frac{1}{(s'\sigma)^2 \circ s^{-1}}.$$

Both recurrence and stationarity are preserved under scale transformation. First, $X$ is recurrent on $\mathcal{D}$ if and only if $X^s$ is recurrent on $\mathbb{R}$. Trivially, the scale function of $X^s$ is identity, since it is already in natural scale, and therefore, $X^s$ is recurrent if and only if its domain is given by the entire real line $\mathbb{R}$. However, the domain of $X^s$ becomes $\mathbb{R}$ if and only if $X$ is recurrent, i.e., $s(\underline{x}) = -\infty$ and $s(\overline{x}) = \infty$. Second, $X$ is stationary on $\mathcal{D}$ if and only if $X^s$ is stationary on $\mathbb{R}$, since $m_s(\mathbb{R}) = m(\mathcal{D})$. Indeed, we may easily show by the change of variables in integrals that $m_s(f_s) = m(f)$ for any $f$ defined on $\mathcal{D}$.

To effectively introduce our asymptotics, we introduce

**Definition 4.2.** Let $f$ be a nonintegrable (or $m$-nonintegrable) regularly varying function on $\mathcal{D}$. We say that $f$ is *strongly nonintegrable* (or $m$-strongly nonintegrable) if $f\ell$ is not integrable (or not $m$-integrable) for any slowly varying function $\ell$. On the other hand, we say that $f$ is *barely nonintegrable* ($m$-barely nonintegrable) if there exists some slowly varying function $\ell$ such that $f\ell$ is integrable (or $m$-integrable).

Following Definition 4.2, we say that a null recurrent diffusion $X$ is *strongly nonstationary* if its speed density $m$ is strongly nonintegrable, and *barely nonstationary* if its speed density $m$ is barely nonintegrable. We assume that $m_s$ and $s^{-1}$ have $+\infty$ as their dominating boundary. This assumption is not restrictive and made just for the convenience of exposition.

Now we provide the fundamental lemma for the mean reversion of diffusion. There we consider two conditions.

(ST) $m$ is either integrable or barely nonintegrable, and
(DD) $\sigma^2$ is either $m$-integrable or $m$-barely nonintegrable.

Clearly, ST is a condition related to the stationarity of $X$, and it holds if and only if $X$ is stationary or barely nonstationary. On the other hand, DD provides a condition that we have a dominating drift in $X$, as will be shown more precisely below. It is easy to see that ST holds if and only if $q \leq -1$ at $x_B = \pm\infty$ and $q \geq -1$ at $x_B = 0$, and DD holds if and only if $p \geq 1$ at $x_B = \pm\infty$. Recall that we assume $p \neq -1$ throughout the paper. Note that recurrence requires $p < -1$ at $x_B = 0$, and therefore, $\sigma^2$ is always $m$-integrable at $x_B = 0$.

We have

**Lemma 4.1.** ST *or* DD *holds if and only if*

$$\int_0^T (X_t - \overline{X}_T)dX_t \sim_p -\frac{1}{2}\int_0^T \sigma^2(X_t)dt$$

*for all large* $T$.

Lemma 4.1 alludes that $X$ has mean reversion if and only if one of the two conditions ST and DD holds. This is indeed true, as will be shown below. Note that we have $(X_t - \overline{X}_T)dX_t = (1/2)\big(d(X_t - \overline{X}_T)^2 - d[X]_t\big)$, due to Itô's formula, and $d[X]_t = \sigma^2(X_t)dt$. Therefore, the only if part of Lemma 4.1 follows straightforwardly, if ST holds with stationary $X$ having all required moments, since in this case $X_T, X_0$ and $\overline{X}_T$ are all of $O_p(1)$, and $(X_T - \overline{X}_T)^2$ and $(X_0 - \overline{X}_T)^2$ become negligible in the limit compared to $[X]_T = \int_0^T \sigma^2(X_t)dt$. Lemma 4.1 shows that the asymptotic negligibility of $(X_T - \overline{X}_T)^2$ and $(X_0 - \overline{X}_T)^2$ is indeed a necessary and sufficient condition for ST or DD.

To better understand our condition DD, we write $(X_t - \overline{X}_T)dX_t = (X_t - \overline{X}_T)\mu(X_t)dt + (X_t - \overline{X}_T)\sigma(X_t)dW_t$. It can be deduced that

**Lemma 4.2.** DD *holds if and only if*

$$\int_0^T (X_t - \overline{X}_T)\sigma(X_t)dW_t \prec_p \int_0^T (X_t - \overline{X}_T)\mu(X_t)dt$$

*for all large* $T$.

Lemma 4.2 shows that DD holds if and only if drift plays a dominating role in mean reversion of a diffusion. Together with Lemma 4.1, Lemma 4.2 implies that DD holds if and only if

$$\int_0^T (X_t - \overline{X}_T)dX_t \sim_p \int_0^T (X_t - \overline{X}_T)\mu(X_t)dt \sim_p -\frac{1}{2}\int_0^T \sigma^2(X_t)dt$$

for all large $T$.

More explicit asymptotics on mean reversion are given in the following theorem. For locally integrable $f$ on $\mathbb{R}$, we define $f[\lambda]$ as

$$f[\lambda] = \int_{|x|<\lambda} f(x)dx.$$

Note in particular that, for $f$ on $\mathcal{D}$, we have

$$(m_s f_s)[\lambda] = \int_{|x|<\lambda} f_s(x)m_s(dx) = \int_{|s(x)|<\lambda} f(x)m(dx),$$

whenever it is well defined. If $f$ is $m$-integrable, we have $(m_s f_s)[\lambda] \to m(f)$ as $\lambda \to \infty$. We let $(\lambda_T)$ be the normalizing sequence satisfying

$$T = \lambda_T m_s[\lambda_T] \quad \text{or} \quad \lambda_T^2 m_s(\lambda_T) \tag{12}$$

depending upon whether or not ST holds, and subsequently define $(C_T)$ from $(\lambda_T)$ as

$$C_T = \begin{cases} \lambda_T (m_s \sigma_s^2)[\lambda_T] & & \text{DD holds} \\ \lambda_T^2 (m_s \sigma_s^2)(\lambda_T) & \text{if} & \text{ST holds and DD does not hold} \\ \left(s^{-1}(\lambda_T)\right)^2 & & \text{neither ST nor DD holds} \end{cases} \tag{13}$$

Furthermore, we let the stopping time $\tau$ be defined as

$$\tau = \inf\left\{ t \,\middle|\, L(t,0) > 1 \right\} \quad \text{or} \quad \inf\left\{ t \,\middle|\, \int_{\mathbb{R}} L(t,x)\overline{m}_s(dx) > 1 \right\} \tag{14}$$

depending upon whether or not ST holds, where $L$ is the local time of Brownian motion $B$. Finally, following our convention, we denote by $\overline{f}$ the limit homogeneous function of regularly varying $f$ on $\mathbb{R}$.

**Theorem 4.3.** *If* DD *holds, we have*

$$\frac{1}{C_T} \int_0^T (X_t - \overline{X}_T) dX_t \to_d -\frac{1}{2} L(\tau, 0).$$

*as $T \to \infty$. On the other hand, if* ST *holds and* DD *does not hold, we have*

$$\frac{1}{C_T} \int_0^T (X_t - \overline{X}_T) dX_t \to_d -\frac{1}{2} \int_0^\tau \overline{m_s \sigma_s^2}(B_t) dt.$$

*as $T \to \infty$.*

Theorem 4.3 provides the details of mean-reverting behavior of diffusion. Both limit distributions in Theorem 4.3 have support on $(-\infty, 0)$, since $L(t,0) > 0$ a.s. for all $t > 0$, and $\int_{|x|<\epsilon} \overline{m_s \sigma_s^2}(x) dx > 0$ for any $\epsilon > 0$ as well as $\tau > 0$ a.s. If ST holds, we have $L(\tau, 0) = 1$ a.s. In fact, if both ST and DD hold, we may easily deduce that $X$ has strong mean reversion. Furthermore, if in particular $m$ is integrable and $\sigma^2$ is $m$-integrable so that $X$ is stationary and $\mathbb{E}\sigma^2(X_t) < \infty$, we have

$$\frac{1}{T} \int_0^T (X_t - \overline{X}_T) dX_t \to_d -\frac{1}{2} \pi(\sigma^2)$$

as $T \to \infty$. Note that $(m_s)[\lambda] \to m(\mathcal{D})$ and $(m_s\sigma_s^2)[\lambda] \to m(\sigma^2)$ as $\lambda \to \infty$, and therefore, $C_T = \pi(\sigma^2)T$, as well as $L(\tau, 0) = 1$ a.s.

If neither ST nor DD holds, we would have a quite different asymptotics. Let $Y = s(X)$ be the scale transformation of $X$ and define $Y^T$ by $Y_t^T = \lambda_T^{-1} Y_{Tt}$ with the normalizing sequence $(\lambda_T)$ in (12), we have $Y^T \to_d Y^\circ$ as $T \to \infty$ in the space $C[0,1]$ of continuous functions with uniform topology, where using Brownian motion $B$ and its local time $L$ we may represent the limit process $Y^\circ$ as

$$Y_t^\circ = B \circ \overline{A}_t \quad \text{with} \quad \overline{A}_t = \inf\left\{ s \,\middle|\, \int_{\mathbb{R}} L(s,x)\overline{m}_s(dx) > t \right\}.$$

To obtain the asymptotics for a general diffusion $X$, we write it as $X = s^{-1}(Y)$ and note that $s^{-1} \in RV_\kappa$ with $\kappa \in (0,2)$ in this case. Therefore, if we define $X^T$ as $X_t^T = X_{Tt}/s^{-1}(\lambda_T) = s^{-1}(Y_{Tt})/s^{-1}(\lambda_T)$, we may well expect that

$$X^T = \frac{s^{-1}(\lambda_T Y^T)}{s^{-1}(\lambda_T)} \to_d \overline{s^{-1}}(Y^\circ) = X^\circ,$$

in $C[0,1]$ as $T \to \infty$.

**Theorem 4.4.** *If neither* ST *nor* DD *holds, we have*

$$\frac{1}{C_T}\int_0^T (X_t - \overline{X}_T)dX_t \to_d \int_0^1 (X_t^\circ - \overline{X}_1^\circ)dX_t^\circ$$

*as $T \to \infty$, where $X^\circ$ is the limit process of $X$ and $\overline{X}_1^\circ = \int_0^1 X_t^\circ dt$.*

The limit distribution in Theorem 4.4 does not have support inside $(-\infty, 0)$, and therefore, we do not have mean reversion if neither ST nor DD holds.

Now it is clear from Theorems 4.3 and 4.4 that diffusion has mean reversion if and only if either ST or DD holds. Moreover, we have strong mean reversion if and only if both ST and DD hold, and weak mean reversion if and only if only one of ST and DD holds.

It is worth noting that the normalizing sequences $(\lambda_T)$ and $(C_T)$ introduced respectively in (12) and (13) can be defined as

$$T = \lambda_T m_s[\lambda_T] \quad \text{and} \quad C_T = \lambda_T (m_s\sigma_s^2)[\lambda_T] \tag{15}$$

in all cases, up to some adjusting constants. Originally, we use them respectively only for the case of ST and DD. For the definition of $(\lambda_T)$ in case that ST does not hold and

$m \in RV_q$ with $q > -1$, we may easily deduce from the Kamarata's theorem that

$$\lambda_T m_s[\lambda_T] = \lambda_T \int_{|x| \le \lambda_T} m_s(x)dx \sim \frac{1}{1+\kappa}\lambda_T^2 m_s(\lambda_T)$$

for $m_s \in RV_\kappa$ with $\kappa > -1$ where $\kappa = (q-p)/(1+p)$ since $m_s = (m/s') \circ s^{-1}$. We may obtain similar results for the definition of $(C_T)$. In case ST holds and DD does not hold, we have

$$\lambda_T (m_s\sigma_s^2)[\lambda_T] = \lambda_T \int_{|x| \le \lambda_T} (m_s\sigma_s^2)(x)dx \sim \frac{1}{1+\kappa}\lambda_T^2 (m_s\sigma_s^2)(\lambda_T)$$

for $m_s\sigma_s^2 \in RV_\kappa$ with $\kappa > -1$ where $\kappa = -2p/(1+p)$ since $m_s\sigma_s^2 = (1/s')^2 \circ s^{-1}$. Finally, we have

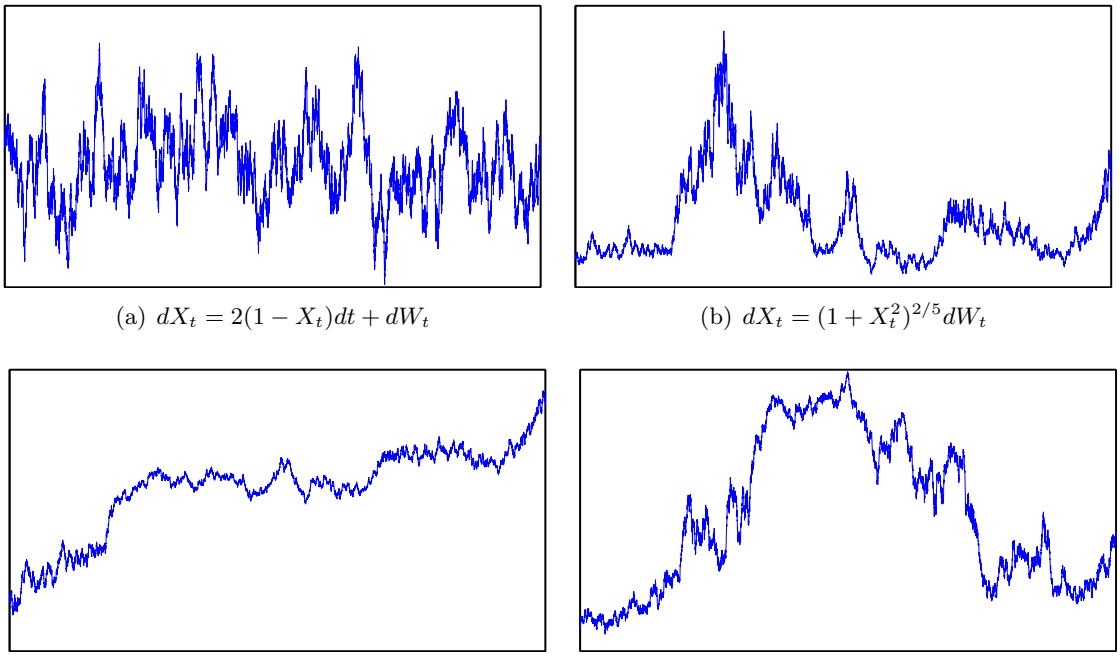
$$\left(s^{-1}(\lambda_T)\right)^2 \sim (1+p)^2 \lambda_T^2 (m_s\sigma_s^2)(\lambda_T) \sim (1-p^2)\lambda_T (m_s\sigma_s^2)[\lambda_T]$$

for $|p| < 1$, since $s^{-1}(\lambda) \sim (1+p)\lambda / \left(s' \circ s^{-1}(\lambda)\right)$ and $(m_s\sigma_s^2)[\lambda] \sim (1+p)/(1-p)\,(\lambda(m_s\sigma_s)(\lambda))$ by Karamata's theorem with $\left(1/s' \circ s^{-1}(\lambda)\right)^2 = (m_s\sigma_s^2)(\lambda)$.

**Example 4.1.** For an illustration, we consider the generalized Höpfner and Kutoyants model (9). As discussed in Example 3.2, we have $p = -2a$ and $q = 2a - 2b$, and therefore, if we set $a < 1/2$, it becomes recurrent on $\mathbb{R}$. Therefore, DD and ST holds if and only if $p \ge 1$ and $q \le -1$, which holds if and only if $a \le -1/2$ and $a - b \le -1/2$ for the diffusion, respectively. Consequently, the diffusion has mean reversion if and only if either $a \le -1/2$ or $a - b \le -1/2$, and if both conditions hold, it has strong mean reversion. However, if $a > -1/2$ and $a - b > -1/2$, the diffusion has no mean reversion.

In Figure 2, we provide the simulated sample paths of four different diffusions with $\Delta = 1/250$ and $T = 40$, which correspond to daily observations for 40 years. The sample path of stationary Ornstein-Uhlenbeck process is in Figure 2 (a). For the Ornstein-Uhlenbeck process, we have $p = \infty$ and $q = -\infty$. Thus it satisfies both ST and DD, and hence, it has strong mean reversion. Figure 2 (b) and (c) present the sample paths of driftless diffusions with $q = -1.6$ and $q = -0.8$, respectively. Recall that all driftless diffusions have $p = 0$. The driftless diffusion in Figure 2 (b) satisfies ST but not DD, and it has weak mean reversion. On the other hand, the driftless diffusion in Figure 2 (c) satisfies neither ST nor DD, and therefore, it is nonstationary and has no mean reversion. Finally, Figure 2 (d) presents the sample path of the generalized Höpfner and Kutoyants model with parameter values $a = -1.4$ and $b = -1$, which yields $p = 2.8$ and $q = -0.8$. The diffusion satisfies DD but not ST, and therefore, it has mean reversion, though it is nonstationary.

**Remark 4.1.** Clearly, we may extend the concept of mean reversion in Definition 4.1 to discrete time series. As an example, we consider the fractionally integrated process $(x_i)$



(a) $dX_t = 2(1 - X_t)dt + dW_t$

(b) $dX_t = (1 + X_t^2)^{2/5} dW_t$

(c) $dX_t = (1 + X_t^2)^{1/5} dW_t$,

(d) $dX_t = -(7/5)X_t(1+X_t^2)^{-2}dt + (1+X_t^2)^{-1/2}dW_t$

Figure 2: Sample Paths of Various Diffusions

with order $-1/2 < d < 3/2$, i.e., $\triangle^d x_i = \varepsilon_i$, where $(\varepsilon_i)$ is an i.i.d. random sequence with mean zero. It is well known that $(x_i)$ is stationary if and only if $-1/2 < d < 1/2$, whereas it is nonstationary if $d \geq 1/2$, see, e.g., Baillie (1996). Under appropriate moment conditions on $(\varepsilon_i)$, it follows from Sowell (1990) and Marmol (1998) that, as $N \to \infty$,

$$\begin{array}{lll} \frac{1}{N}\sum_{i=1}^N (x_{i-1} - \overline{x}_N)\triangle x_i \to_p (2\delta - 1/1 - \delta)\mathbb{E}u_i^2 < 0, & & -1/2 < d < 1/2 \\ \frac{1}{N}\sum_{i=1}^N (x_{i-1} - \overline{x}_N)\triangle x_i \to_p -\mathbb{E}u_i^2/2 < 0, & & 1/2 < d < 1 \\ \frac{1}{\sigma_N^2}\sum_{i=1}^N (x_{i-1} - \overline{x}_N)\triangle x_i \to_d \int_0^1 (W_t - \overline{W}_1)dW_t, & \text{if} & d = 1 \\ \frac{1}{\sigma_N^2}\sum_{i=1}^N (x_{i-1} - \overline{x}_N)\triangle x_i \to_d W_1^{\delta 2}/2 - W_1^\delta \overline{W}_1^\delta > 0, & & d > 1 \end{array}$$

where $u_i = \triangle^\delta \varepsilon_i$ with $\delta = 1 - d$, $\sigma_N^2 = \text{var}\left(\sum_{i=1}^N u_i\right)$ and $W^\delta$ is the fractional Brownian motion defined as $W_t^\delta = (1/\Gamma(1+\delta)) \int_0^t (t-s)^\delta dW_s$. Consequently, $(x_i)$ has mean reversion if and only if $-1/2 < d < 1$. For $d \geq 1$, $(x_i)$ has no mean reversion, and if $d > 1$, in particular, $(x_i)$ has anti-mean reversion.

# 5 Unit Root Test

In the section, we consider testing for a unit root in discrete samples from diffusion models. More precisely, we let the discrete sample

$$(X_\Delta, ..., X_{i\Delta}, ..., X_{N\Delta})$$

of size $N$ be collected from the diffusion process $X$ defined in (1) at times $\Delta, \ldots, N\Delta$ over interval $[0, T]$ with $T = N\Delta$, and test where there is a unit root in the sample. For the brevity of notation, we will simply write

$$x_i = X_{i\Delta}$$

for all $i = 1, \ldots, n$ with $x_0 = X_0$.

To test for a unit root in $(x_i)$, we may use the first-order autoregression without any augmented lags, since $X$ is a markov process. In particular, we consider the regression

$$\triangle x_i = \alpha + \beta x_{i-1} + \varepsilon_i, \tag{16}$$

where $\triangle$ is the usual difference operator, and test the null hypothesis $\beta = 0$ against the alternative hypothesis $\beta < 0$ using the least squares regression. The least square estimator and the $t$-statistic for $\beta$ in (16), denoted respectively as $\hat{\beta}$ and $t(\hat{\beta})$, are given by

$$\hat{\beta} = \frac{\sum_{i=1}^{N}(x_{i-1} - \overline{x}_N)\triangle x_i}{\sum_{i=1}^{N}(x_{i-1} - \overline{x}_N)^2}$$

and

$$t(\hat{\beta}) = \frac{\hat{\beta}}{\hat{\sigma}\left(\sum_{i=1}^{N}(x_{i-1} - \overline{x}_N)^2\right)^{-1/2}},$$

where $\overline{x}_N$ is the sample mean of $(x_i)$ and $\hat{\sigma}^2$ is the usual estimator for the variance of regression errors $(\varepsilon_i)$.

In the usual discrete time setup, the Dickey-Fuller test based on the $t$-statistic $t(\hat{\beta})$ from regression (16) is widely used to test for the null hypothesis of a unit root, i.e., $\beta = 0$, against the alternative hypothesis of stationarity, i.e., $\beta < 0$. Under the null hypothesis of a unit root, it has nonstandard, yet well-defined, limit distribution, as long as some mild regularity conditions are satisfied for the innovations $(\varepsilon_i)$. The limit distribution, which we call the Dickey-Fuller distribution, is usually represented as a functional of Brownian motion. On the other hand, under the alternative hypothesis of stationarity, it diverges to negative

infinity in probability. Consequently, it provides a test for unit root nonstationarity that is consistent against a wide class of stationary time series. Therefore, we may say, loosely yet generally, that a unit root time series is nonstationary, and that a stationary time series does not have a unit root.

The Dickey-Fuller test has also routinely been applied to samples that are thought to be collected discretely from diffusion type continuous time processes. However, little is known about its asymptotic behavior for discrete samples obtained from general continuous time processes. In particular, except for simple models such as Brownian motion and Ornstein-Uhlenbeck process, it has been completely unknown whether the test can be used to effectively discriminate nonstationary diffusions from stationary diffusions. To facilitate our discussions on the asymptotic behavior of the Dickey-Fuller test, we will simply say in what follows that a diffusion has a unit root if and only if $t(\hat{\beta})$ is stochastically bounded and the test is expected to suggest the non-rejection of the unit root hypothesis at least with some positive probability.

## 5.1 Continuous Time Approximation of Unit Root Test

In this section, we present the primary asymptotics for the unit root test. We start with some technical assumptions. We let $\iota$ denote the identity function, i.e., $\iota(x) = x$.

**Assumption 5.1.** *$\iota, \mu, \sigma$ are all majorized by $\omega : \mathcal{D} \to \mathbb{R}$.*

In Assumption 5.1, we require the existence of an envelop function for $\mu$, $\sigma$ and $\iota$ so that we may effectively control them near the boundaries. Indeed, the envelop function $\omega$ always exists since we may simply set $\omega(x) = \max\{\mu(x), \sigma(x), x\}$.

In our asymptotics, we require that the sampling interval $\Delta$ be sufficiently small relative to the extremal bounds of various functional transforms of $X$ over time interval $[0, T]$. Following Aït-Sahalia and Park (2011), we define

$$T(f) = \max_{0 \le t \le T} |f(X_t)|$$

for some function $f : \mathcal{D} \to \mathbb{R}$. For the identity function, we have $T(\iota) = \max_{0 \le t \le T} |X_t|$ and $T(\iota)$ becomes the asymptotic order of extremal process of $X$. It is obvious that we have $[T(\iota)]^k = T(\iota^k)$ for any nonnegative $k$. More generally, for any regularly varying $f$, we may obtain the exact rate of $T(f)$ from the asymptotic behavior of extremal process. For instance, the extremal processes of Ornstein-Uhlenbeck process and Feller's square root process are respectively of orders $O_p(\sqrt{\log T})$ and $O_p(\log T)$, and the extremal process of the general driftless diffusion process is of order $O_p(T)$. Thus if $f$ is regularly varying and $c_T$ is

the order of extremal process, then we have $T(f) = f(c_T)$. The order of extremal process is known for a wide class of diffusions. For instance, under some regularity conditions on $\mu$ and $\sigma^2$, it is well known that the extremal processes of positive recurrent diffusions are of order $O_p(s^{-1}(T))$, to which the reader is referred to, e.g., Davis (1982). Moreover, asymptotic orders for the extremal processes of general null recurrent diffusions are obtained by Stone (1963), Jeong and Park (2011) and Kim and Park (2013).

**Assumption 5.2.** $\Delta^{1/2}T(\omega^2)T\sqrt{\log 1/[\Delta T(\omega^2)]} \to 0$.

Assumption 5.2 makes it necessary to have $\Delta \to 0$. If we fix $T$, $\Delta \to 0$ is indeed the necessary and sufficient condition for Assumption 5.2. Clearly, we may also allow $T \to \infty$ as long as $\Delta \to 0$ sufficiently fast. In this case, our asymptotic results will be more relevant for the case where $\Delta$ is sufficiently small relative to $T$. Our asymptotics in the paper are derived under the condition $\Delta \to 0$ and $T \to \infty$ jointly. For Assumption 5.2 to hold, it suffices to have $\Delta = O(T^{-2-\epsilon})$ for any $\epsilon > 0$, if $X$ is bounded so that $T(\omega^2)$ is a constant. The condition appears to be mild enough to yield asymptotics generally relevant for a very wide range of empirical analysis relying on samples collected from diffusion models. For daily observations over ten years, as an example, we have $\Delta \approx 1/250$ and $T^{-2} = 1/100$. Our subsequent asymptotics hold jointly in $\Delta$ and $T$ as long as they satisfy Assumption 5.2 as $\Delta \to 0$ and $T \to \infty$. In particular, we do not use sequential asymptotics, requiring $\Delta \to 0$ and $T \to \infty$ sequentially.

The primary asymptotics for $\hat{\beta}$ and $t(\hat{\beta})$ are given by the following lemma.

**Lemma 5.1.** *Let Assumption 5.1 and 5.2 hold. We have*

$$T\Delta^{-1}\hat{\beta} \sim_p \frac{T\int_0^T (X_t - \overline{X}_T)dX_t}{\int_0^T (X_t - \overline{X}_T)^2 dt},$$
$$t(\hat{\beta}) \sim_p \frac{\sqrt{T}\int_0^T (X_t - \overline{X}_T)dX_t}{[X]_T^{1/2}\left(\int_0^T (X_t - \overline{X}_T)^2 dt\right)^{1/2}}$$

*for all $\Delta$ sufficiently small relative to $T$.*

The limit theory in Lemma 5.1 holds for all small enough $\Delta$ relative to $T$ as long as $\Delta$ and $T$ satisfy Assumption 5.2. Therefore, we expect that they provide good approximations for finite sample distributions, whenever $\Delta$ is relatively small compared with $T$. Note that we do not assume $T = \infty$ to obtain the asymptotics in Lemma 5.1. The asymptotics we have in Lemma 5.1 will be referred in the paper to as the primary asymptotics. The joint asymptotics for $\Delta \to 0$ and $T \to \infty$, which are presented below, may be obtained simply by taking $T$-limits to our primary asymptotics.

Our primary asymptotics in Lemma 5.1 make it clear that we have $\hat{\beta} \to_p 0$ whenever $\Delta \to 0$ fast enough relative to $T$. If, in particular, $T$ is fixed, we have $\hat{\beta} \to_p 0$ for any diffusion $X$ as long as $\Delta \to 0$. For discrete samples from any diffusion, we will therefore always observe a root getting close to unity as we collect samples frequently enough and $\Delta$ becomes sufficiently small. However, even in this case, the unit root test will not necessarily support the presence of a unit root. If we let $\Delta \to 0$ with fixed $T$, $t(\hat{\beta})$ remains to be random and the unit root test will yield completely random conclusion, regardless of the asymptotic properties of the underlying diffusion. The unit root test will be totally uninformative in this case. Indeed, it is very natural and not surprising at all that the unit root test does not yield any information on the asymptotics, such as mean reversion and unit root properties, of the underlying diffusion unless $T$ becomes large. In discussions to follow, we let $T \to \infty$ and consider large $T$ asymptotics.

## 5.2 Asymptotics of Unit Root Test

Now we establish the asymptotics for the unit root test.

**Theorem 5.2.** *Let Assumption 5.1 and 5.2 hold. If neither* ST *nor* DD *holds, we have*

$$T\Delta^{-1}\hat{\beta} \to_d \frac{\int_0^1 (X_t^\circ - \overline{X}_1^\circ) dX_t^\circ}{\int_0^1 (X_t^\circ - \overline{X}_1^\circ)^2 dt},$$
$$t(\hat{\beta}) \to_d \frac{\int_0^1 (X_t^\circ - \overline{X}_1^\circ) dX_t^\circ}{[X^\circ]_1^{1/2} \left( \int_0^1 (X_t^\circ - \overline{X}_1^\circ)^2 dt \right)^{1/2}}$$

*as* $\Delta \to 0$ *and* $T \to \infty$. *On the other hand, if either* ST *or* DD *holds, we have*

$$T\Delta^{-1}\hat{\beta} \to_p -\infty \quad \textit{and} \quad t(\hat{\beta}) \to_p -\infty$$

*as* $\Delta \to 0$ *and* $T \to \infty$.

Theorem 5.2 shows that $N\hat{\beta}$ and $t(\hat{\beta})$ have nondegenerate limit distributions if and only if neither ST nor DD holds. If either ST or DD holds, $N\hat{\beta} \to_p -\infty$ and $t(\hat{\beta}) \to_p -\infty$. Note in particular that $T\Delta^{-1} = N$. In other words, the normalized estimator of $\beta$ and its $t$-ratio, the two most commonly used statistics for the unit root, may not reject the unit root hypothesis even in the limit if and only if neither ST nor DD holds. If either ST or DD holds, both statistics are expected to reject the unit root hypothesis with probability one in the limit. Clearly, the necessary and sufficient condition we require here to have the evidence of a unit root from the unit root test is precisely the same as the necessary and

| $T$ | 10 | 20 | 40 | 80 |
|---|---|---|---|---|
| Rejection Probabilities | 10.1% | 14.2% | 20.3% | 29.9% |

Table 1: Rejection probabilities of Dickey-Fuller test for nonstationary diffusion having dominating drift

sufficient condition for the absence of mean reversion. That is, we have a unit root and the unit root test supports the presence of a unit root, if and only if the underlying diffusion does not have mean reversion. In particular, the test of a unit root in discrete samples from diffusion does not provide any clue for, or against, the nonstationarity of the underlying diffusion.

It is clear from Theorem 5.2 that the unit root test cannot be used to test for nonstatonarity. Null recurrent diffusion will not have a unit root if DD holds and it has mean reversion due to the dominant role of drift in its longrun behavior. As an illustrative example, we consider the generalized Höpfner and Kytoyants model in (9) with $a = -7/5$ and $b = -1$. With the given parameter values, $X$ has a dominating drift though it is nonstationary, i.e., DD holds though ST is not satisfied. Therefore, it is mean-reverting and the unit root test is expected to reject the unit root hypothesis. To demonstrate this, we compute the actual rejection probabilities of the unit root test and report them in Table 1. The reported rejection probabilities are obtained by simulating 10,000 realizations of the diffusion with $\Delta = 1/250$ and $T = 10, 20, 40, 80$, which correspond to daily observations for the periods of 10, 20, 40, 80 years, respectively. We use the 5% critical value from the Dickey-Fuller distribution. As expected, the actual rejection probabilities exceed the nominal level for all $T$, and they increase with $T$.

We should also note that the unit root test does not have any nontrivial power in discriminating diffusions with and without drift. As we have seen earlier, recurrent diffusion with linear drift is positive recurrent. Therefore, as long as it is recurrent, any diffusion is stationary if it has a linear drift. This, in turn, implies that the unit root test rejects the unit root hypothesis for any recurrent diffusion with linear drift. However, in general, the unit root test does not have any discriminatory power for or against the presence of drift in diffusion. It is clear that for a driftless diffusion may also be mean-reverting, and has no unit root, as long as it is stationary or barely nonstationary.

The limit distribution for the unit root test $t(\hat{\beta})$ is model-dependent. In particular, its limit distribution is generally different from the Dickey-Fuller distribution and the standard critical values for the unit root test cannot be used. For an illustration of model dependency of the limit distribution of the unit root test, we consider the generalized Höpfner and

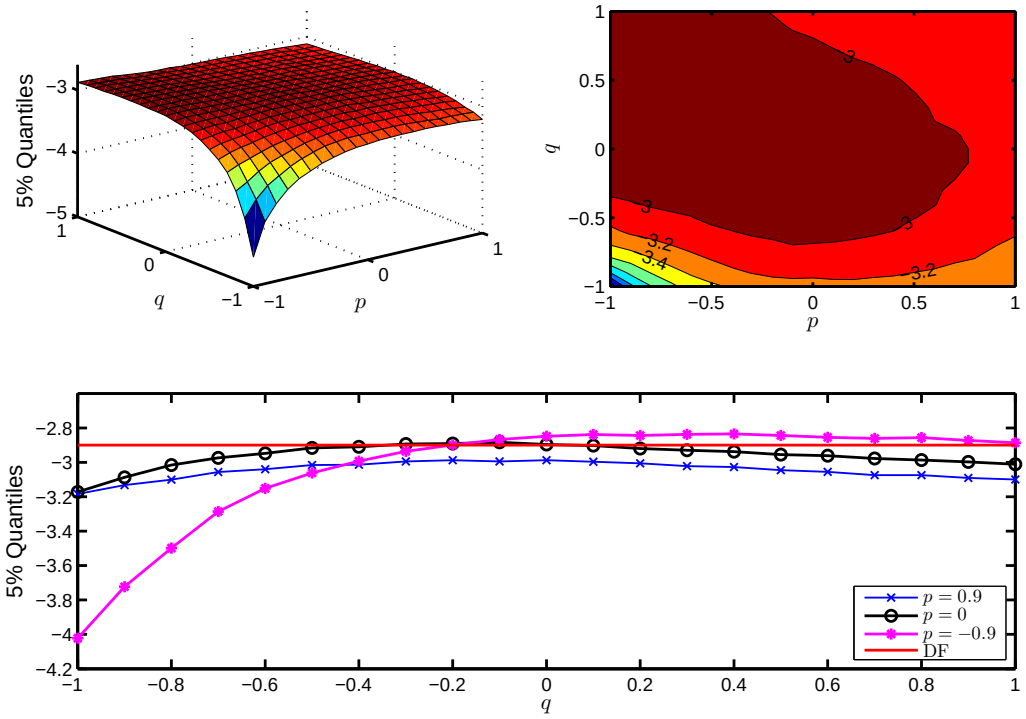



Figure 3: Illustration of the model dependency in limiting distributions under nonstationarity. The data were generated according to generalized Höpfner and Kutoyants model (9). The plot on the upper-left presents 5% quantiles of $t$-statistics for $(p, q) \in [-1, 1] \times [-1, 1]$, and its corresponding Contour plot is on the upper-right panel. In the plot on the lower panel, we compare the critical value from the Dickey-Fuller distribution with 5% quantiles of $t$-statistics for $p = -0.9, 0, 0.9$ and $q \in [-1, 1]$.

Kutoyants model in (9). For the generalized Höpfner and Kutoyants model, we have $p = -2a$ and $q = 2a - 2b$. As shown in Theorem 5.2 (a), different choice of $(p, q)$ gives different limiting distributions. We conduct simulations for the process (9) with various combinations of $(p, q) \in [-1, 1] \times [-1, 1]$ to examine the model dependency in null distributions. For each experiment, we simulate 10,000 realizations with $\Delta = 1/250$ and $T = 40$ which correspond to daily data for 40 years. Figure 3 shows the 5% quantiles of $t$-statistics. There appears to be considerable amount of model dependency in the finite sample distribution. As we can see in Figure 3, the critical values from the Dickey-Fuller distributions may give misleading conclusions. In particular, if both $p$ and $q$ are close to, but not less than, -1, then 5% quantiles of $t$-statistics from (9) are significantly smaller than the critical value from the Dickey-Fuller distributions. For example, if $p = -0.9$ and $q = -0.9$, then the 5% quantile of $t$-statistics from (9) is about $-3.7$, whereas corresponding critical value from the Dickey-Fuller distributions is about $-2.9$.

Of course, our limit distribution of the unit root test reduces to the Dickey-Fuller distribution in case the limit process $X^\circ$ of the underlying diffusion $X$ becomes Brownian motion. Clearly, $X^\circ$ becomes Brownian motion if and only if $\overline{\sigma^2}$ and $\overline{s'}$ are constant functions. Therefore, we need to have $p = 0$ and $q = 0$ at both boundaries. Moreover, it is required that $\sigma^2(-\lambda)/\sigma^2(\lambda) \to 1$, and

$$\frac{s'(-\lambda)}{s'(\lambda)} = \exp\left(-\int_{|x|\le\lambda} \frac{2\mu(x)}{\sigma^2(x)}dx\right) \to 1,$$

which holds if and only if

$$\int_{|x|\le\lambda} \frac{\mu(x)}{\sigma^2(x)}dx \to 0,$$

as $\lambda \to \infty$. For instance, the diffusion defined as

$$dX_t = \frac{X_t}{1+X_t^2}dt + \big(1+\log^{1/2}(1+|X_t|)\big)dW_t$$

becomes Brownian motion in the limit.

If used to test for nonstationarity of the underlying diffusion, the unit root test is consistent against all mean-reverting diffusions. However, the test becomes consistent and has discriminating power only when the time span $T$ goes to infinity. In particular, it is well expected from our theory that the test will not be consistent if we just increase the sample size $N = T/\Delta$ by decreasing the sampling interval $\Delta$ (or equivalently, increasing sampling frequency $1/\Delta$) with fixed $T$. This was first observed in Shiller and Perron (1985), and further analyzed later by Perron (1989, 1991), for the test of a unit root in samples from Brownian motion against those from Ornstein-Uhlenbeck process.

**Remark 5.1.** In the paper, we mainly consider stock variables, for which we assume that the discrete samples $(x_i)$ are obtained from the underlying diffusion at equally spaced time intervals and let $x_i = X_{i\Delta}$. However, we may also analyze the discrete samples for flow variables, which are given as

$$x_i = \int_{(i-1)\Delta}^{i\Delta} X_t dt$$

from the underlying diffusion $X$. The asymptotics of the unit root test for the discrete samples from continuous time flow variables is investigated by Chambers (2004, 2008), under the assumption that they are generated by either Brownian motion or Ornstein-Uhlenbeck process. For some related simulation studies of the unit root test for the samples from continuous time processes, the reader is referred to Choi (1992), Choi and Chung (1995) and Boswijk and Klaassen (2012).

**Remark 5.2.** Let $(x_i)$ be a fractionally integrated process with order $d$ introduced in Remark 4.1. Under appropriate moment conditions on $(\varepsilon_i)$, Sowell (1990) and Marmol (1998) show that $t(\hat{\beta}) \to_p -\infty$ if $d < 1$, and $t(\hat{\beta}) \to_p \infty$ if $d > 1$, as $N \to \infty$. As is well known, if $d = 1$, the $t$-ratio converges in distribution to the Dickey-Fuller distribution as $N \to \infty$. Therefore, we may use the unit root test to investigate the presence of mean reversion in fractionally integrated processes.

# 6 Conclusion

In this paper, we study the longrun behaviors of general diffusion models and find many interesting results. In particular, we show that the Dickey-Fuller statistic cannot be used to test for nonstationarity of the underlying diffusion. Rather, it becomes a test for no mean reversion of the underlying diffusion, if applied to discrete samples from a diffusion. A diffusion has a unit root in its discrete samples if and only if it does not have mean reversion, and a nonstationary diffusion may also be mean-reverting if it has a drift dominating its diffusion. The test would at least have the perfect discriminatory power against all stationary diffusions, if we only consider nonstationary and non-mean-reverting diffusions under the null hypothesis. However, even in this case, the limit distribution of the test becomes model-dependent and the usual Dickey-Fuller critical values are not applicable. For the unit root test applied to discrete samples from a diffusion model to be strictly valid as a test for nonstationarity of the underlying diffusion, therefore, we should further restrict the class of diffusion models considered under the null hypothesis to be nonstationary diffusions that become Brownian motions in the limit.

# Appendix A Useful Lemmas

In the following, we assume that $X$ is recurrent diffusion satisfying Assumption 2.1, 3.1 and 3.2. Thus $s' \in RV_p$ with $p \neq -1$, and $m \in RV_q$. Moreover, we let $X^s$ be the scale transformed process of $X$, i.e., $X_t^s = s(X_t)$ for all $t \geq 0$, and denote $m_s$ by the speed measure of $X^s$, which is given by $m_s = 1/(s'\sigma)^2 \circ s^{-1}$.

To facilitate our exposition, we make some notational conventions. For locally integrable $f$ on $\mathbb{R}$, we define $f[\lambda] = \int f(x)1\{|x| < \lambda\}dy$. Note that if $f$ is either integrable or barely nonintegrable, then $f[\lambda]$ is slowly varying. Moreover, we let $(\lambda_T)$ and $(d_T)$ be the normalizing sequence satisfying $d_T = s^{-1}(\lambda_T)$ and

$$T = \begin{cases} \lambda_T m_s[\lambda_T], & \text{if ST holds} \\ \lambda_T^2 m_s(\lambda_T), & \text{otherwise.} \end{cases}$$

Recall that $\lambda_T \sim Tm(\mathcal{D})$ if $X$ is stationary, whereas $\lambda_T = o(T)$ if $X$ is nonstationary. Using $\lambda_T$, we define $T_f$ by

$$T_f = \begin{cases} \lambda_T(m_s f_s)[\lambda_T], & \text{if } f \text{ is either } m\text{-I or } m\text{-BN} \\ \lambda_T^2(m_s f_s)(\lambda_T), & \text{if } f \text{ is } m\text{-SN.} \end{cases}$$

where we mean I, BN and SN by integrable, barely nonintegrable and strongly nonintegrable, respectively. Sometimes, we write $f[a,b] = \int_a^b f(x)dx$ for locally integrable function $f$.

**Lemma A1.** *(a) Let $f$ and $g^2$ be m-I or m-SN. Then we have*

$$\frac{1}{T_f}\int_0^T f(X_t)dt \to_d \begin{cases} L(\tau,0), & \textit{if } f \textit{ is m-I} \\ \int_0^\tau \overline{m_s f_s}(B_t)dt, & \textit{if } f \textit{ is m-SN} \end{cases}$$

*and*

$$\frac{1}{\sqrt{T_{g^2}}}\int_0^T g(X_t)dW_t \to_d \begin{cases} \sqrt{L(\tau,0)}N, & \textit{if } g^2 \textit{ is m-I} \\ \int_0^\tau \overline{m_s^{1/2} g_s}(B_t)dB_t, & \textit{if } g^2 \textit{ is m-SN} \end{cases}$$

*where $B$ is Brownian motion with local time $L$, $N$ is standard normal variate independent of $B$, and $\tau$ is time change defined as*

$$\tau = \begin{cases} \inf\{s|L(s,0) > 1)\}, & \textit{if ST holds} \\ \inf\left\{s|\int \overline{m}_s(x)L(s,x) > 1\right\}, & \textit{otherwise.} \end{cases}$$

*(b) Let $f$ be $m$-BN. Then*

$$\frac{1}{T_{|f|}}\int_0^T f(X_t)dt \to_d KL(\tau,0)$$

*where $K=\lim_{\lambda\to\infty}(m_sf_s)[\lambda]/|m_sf_s|[\lambda]$, and $L$ and $\tau$ are defined in part (a). In particular, if $\lambda(m_sf_s)(\lambda)\prec(m_sf_s)[\lambda]$, then*

$$\frac{1}{T_f}\int_0^T f(X_t)dt \to_d L(\tau,0)$$

*where $L$ and $\tau$ are defined in part (a).*

*(c) Let $f$ be $m$-I with $m(f)=0$. If $(mf)[\underline{x},x]$ is square integrable, then*

$$\frac{1}{\sqrt{\lambda_T}}\int_0^T f(X_t)dt \to_d \left(4L(\tau,0)\int_{\mathcal{D}}\{(mf)[\underline{x},x]\}^2\,dx\right)^{1/2} N$$

*where $N$ is standard normal variate independent of Brownian local time $L$.*

*(d) If ST holds, then $L(\tau,0)=1$ with probability one.*

*Proof for Lemma A1.* The statements in the lemma follows from Kim and Park (2013), and we omit the proof here. □

**Lemma A2.** *(a) If ST holds, then $\overline{X}_T=o_p(d_T)$ and $X_T=o_p(d_T)$ for large $T$.*

*(b) If DD holds, then $\overline{X}_T=o_p(\sqrt{\lambda_T(m_s\sigma_s^2)[\lambda_T]})$ for large $T$.*

*Proof for Lemma A2 (a).* For the first statement, we note that $\overline{X}_T=O_p(T_{|\iota|}/T)$ due to Lemma A1, where $T_{|\iota|}$ is determined by the $m$-integrability of $\iota$. Thus it suffice to show that $T_{|\iota|}/T=o(d_T)$.

Trivially, if $\iota$ is $m$-I, then $T_{|\iota|}/T=o(d_T)$ since $T_{|\iota|}\sim\lambda_T m(|\iota|)=O(T)$ and $d_T\to\infty$. On the other hand, if $s'$ is rapidly varying, then $m$ is rapidly decreasing, and hence, $\iota$ is always $m$-I. In other word, if $\iota$ is $m$-non integrable, then $s'$ is regularly varying, which implies that $s^{-1}$ is regularly varying but not slowly varying. Thus if $\iota$ is $m$-BN, then $s^{-1}(\lambda_T)/\lambda_T^{\epsilon}\to\infty$ for some $\epsilon>0$, and therefore, we have $T_{|\iota|}/T=o(d_T)$ because

$$\frac{T_{|\iota|}}{d_TT}=\frac{|m_s\iota_s|[\lambda_T]}{s^{-1}(\lambda_T)m_s[\lambda_T]}\le\frac{\ell(\lambda_T)}{s^{-1}(\lambda_T)}\to 0$$

for some slowly varying $\ell$, since both $m_s[\lambda]$ and $|m_s\iota_s|[\lambda]$ are slowly varying.

Now let $\iota$ be $m$-SN. Note that if $m$ is I on $\mathcal{D}$, then $m_s[\lambda_T]\to m(\mathcal{D})$ and $\lambda_Tm_s(\lambda_T)\to 0$ since $m_s$ is I on $\mathbb{R}$. Moreover, if $m$ is BN, then $\lambda_Tm_s(\lambda_T)/m_s[\lambda_T]\to 0$ by Karamata's

theorem for barely nonintegrable function (see, e.g., Proposition 1.5.9 in Bingham et al. (1993)). In both cases, we have

$$\frac{T_{|\iota|}}{d_T T} = \frac{\lambda_T (m_s \iota_s)(\lambda_T)}{s^{-1}(\lambda_T) m_s[\lambda_T]} = \frac{\lambda_T m_s(\lambda_T)}{m_s[\lambda_T]} \to 0 \tag{A.1}$$

since $s^{-1} = \iota_s$, which complete the proof for $\overline{X}_T = o_p(d_T)$.

The second statement $X_T = o_p(d_T)$ follows immediately from Kim and Park (2013), and the proof is therefore omitted here. □

*Proof for Lemma A2 (b).* In the proof, we only consider the case where $\iota$ is $m$-SN because the results for $m$-I or $m$-BN $\iota$ can be shown in a similar way used in the proof of part (a).

Now let $\iota$ be $m$-SN. If ST holds, then $\overline{X}_T = o_p(d_T)$ due to Lemma A2 (a). On the other hand, if ST does not hold, then we can deduce from Lemma A1 that $\overline{X}_T = O_p(d_T)$ because we have $d_T = T_\iota / T$ since $T_\iota = \lambda_T^2 (m_s \iota_s)(\lambda_T)$, $d_T = s^{-1}(\lambda_T) = \iota_s(\lambda_T)$ and $T = \lambda_T^2 m_s(\lambda_T)$. In all cases, we have $\overline{X}_T = O_p(d_T)$. Thus it suffice to show that $d_T / \sqrt{\lambda_T (m_s \sigma_s^2)[\lambda_T]} \to 0$ when DD holds.

Note that

$$(m_s \sigma_s^2)[\lambda_T] = \int_{|x|<\lambda_T} \frac{1}{(s' \circ s^{-1}(x))^2} dx = \int_{|x|<d_T} \frac{1}{s'(x)} dx = (1/s')[d_T]$$

which is slowly varying satisfying $\lambda / \left(s'(\lambda)(1/s')[d_T]\right) \to 0$ due to Karamata's theorem for integrable or barely nonintegrable function. Thus

$$\frac{d_T}{\sqrt{\lambda_T (m_s \sigma_s^2)[\lambda_T]}} = \frac{d_T}{\sqrt{s(d_T)(1/s')[d_T]}} \sim \left( \frac{(1+p) d_T}{s'(d_T)(1/s')[d_T]} \right)^{1/2} \to 0 \tag{A.2}$$

since $s(\lambda) \sim \lambda s'(\lambda)/(1+p)$ by Karamata's theorem for strongly nonintegrable function, which complete the proof. □

**Lemma A3.** *(a) If DD holds, then*

$$\overline{X}_T \int_0^T \mu(X_t) dt \prec_p \int_0^T X_t \mu(X_t) dt$$

*for large $T$.*

*(b) If DD does not hold, then*

$$\overline{X}_T \int_0^T \mu(X_t) dt \precsim_p \int_0^T X_t \mu(X_t) dt$$

*for large $T$.*

*Proof for Lemma A3 (a).* Since $m(\sigma^2) = \int_{\mathcal{D}} 1/s'(x)dx$, $\sigma^2$ is $m$-I or $m$-BN if and only if $1/s'$ is I or BN, respectively. Moreover, DD holds if and only if $p \geq 1$ at $x_B = \pm\infty$. Here, we don't need to consider the $m$-integrability of $\sigma^2$ around the zero boundary because $\sigma^2$ is always integrable around zero boundary because $(m\sigma^2)[\underline{x}, x] = \int_0^x 1/s'(y)dy < \infty$ for any $x \in \mathcal{D}$ since $p < -1$ at $x_B = 0$. In the following, we will consider each of $m$-I and $m$-BN cases separately.

**($m$-I $\sigma^2$)** Note that if $\sigma^2$ is $m$-I, then $x/s'(x) \to 0$ as $x \to \overline{x}$ and $\underline{x}$ since $m(\sigma^2) = \int_{\mathcal{D}} 1/s'(x)dx < \infty$ and $s'$ is regularly varying due to Assumption 3.1. Then $\iota\mu$ is also $m$-I and satisfies $m(\sigma^2) = -2m(\iota\mu)$ because

$$m(\sigma^2) = \int_{\mathcal{D}} \frac{1}{s'(x)}dx = \int_{\mathcal{D}} \frac{xs''(x)}{(s'(x))^2}dx + \left[\lim_{x\to\overline{x}} \frac{x}{s'(x)} - \lim_{x\to\underline{x}} \frac{x}{s'(x)}\right] = -2m(\iota\mu)$$

due to the integration by parts with $s''(x) = -2\mu(x)s'(x)/\sigma^2(x)$. Thus it follows from Lemma A1 that

$$\frac{1}{\lambda_T}\int_0^T X_t\mu(X_t)dt \to_d m(\iota\mu)L(\tau, 0) = -\frac{1}{2}m(\sigma^2)L(\tau, 0) \tag{A.3}$$

where $L$ and $\tau$ are defined in Lemma A1.

Moreover, $\sigma^2$ is $m$-I implies that $\mu$ is $m$-I and $m(\mu) = 0$ since

$$m(\mu) = \int_{\mathcal{D}} \frac{\mu(x)}{\sigma^2(x)s'(x)}dx = -\frac{1}{2}\int_{\mathcal{D}} \frac{s''(x)}{(s'(x))^2}dx = \frac{1}{2}\left[\lim_{x\to\overline{x}} \frac{1}{s'(x)} - \lim_{x\to\underline{x}} \frac{1}{s'(x)}\right] = 0.$$

It is easy to see that $s''$ is regularly varying function by Assumption 3.1 with the representation theorem for regularly varying function. Then we have

$$(m\mu)[\underline{x}, x] = \int_{\underline{x}}^x m\mu(y)dy = m(\mu) - \int_x^{\overline{x}} m\mu(y)dy = \frac{1}{2}\int_x^{\overline{x}} \frac{s''(y)}{(s'(y))^2}dy \sim -\frac{p}{2s'(x)}$$

due to Karamata's theorem for integrable function (see, e.g., Proposition 1.5.10 in Bingham et al. (1993)). Therefore, $(m\mu)[\underline{x}, x]$ is square integrable since $s' \in RV_p$ with $p \geq 1$, and hence, we have

$$\int_0^T \mu(X_t)dt = O_p(\sqrt{\lambda_T}). \tag{A.4}$$

by Lemma A1 (c). The stated result for $m$-I $\sigma^2$ follows immediately from (A.3) and (A.4) with Lemma A2 (b).

**($m$-BN $\sigma^2$)** We note that $\iota\mu$ is $m$-BN and $(m_s\sigma_s^2)[\lambda] \sim -2(m_s\iota_s\mu_s)[\lambda]$ as $\lambda \to \infty$ because

$$-2(m_s\iota_s\mu_s)[\lambda] = -2\int_{-\lambda}^{\lambda} m_s\iota_s\mu_s(x)dx = -2\int_{s^{-1}(-\lambda)}^{s^{-1}(\lambda)} m\iota\mu(x)dx = \int_{s^{-1}(-\lambda)}^{s^{-1}(\lambda)} \frac{xs''(x)}{[s'(x)]^2}dx$$
$$= \int_{s^{-1}(-\lambda)}^{s^{-1}(\lambda)} \frac{1}{s'(x)}dx - \left[\frac{s^{-1}(\lambda)}{s' \circ s^{-1}(\lambda)} - \frac{s^{-1}(-\lambda)}{s' \circ s^{-1}(-\lambda)}\right]$$

due to the integration by parts, and

$$\lim_{\lambda\to\infty} \frac{-2(m_s\iota_s\mu_s)[\lambda]}{(m_s\sigma_s^2)[\lambda]} = 1 - \lim_{\lambda\to\infty} \frac{1}{(m_s\sigma_s^2)[\lambda]}\left[\frac{s^{-1}(\lambda)}{s' \circ s^{-1}(\lambda)} - \frac{s^{-1}(-\lambda)}{s' \circ s^{-1}(-\lambda)}\right] = 1 \tag{A.5}$$

by Karamata's theorem for barely nonintegrable function. Moreover, we have $\lambda_T(m_s\iota_s\mu_s)(\lambda_T) \prec (m_s\iota_s\mu_s)[\lambda_T]$ because $d_T/s'(d_T) \prec -2(m_s\iota_s\mu_s)[\lambda_T]$ due to (A.5), and

$$\lambda_T(m_s\iota_s\mu_s)(\lambda_T) = \lambda_T\frac{(m\iota\mu)}{s'} \circ s^{-1}(\lambda_T) \sim \frac{-2p\lambda_T}{(s')^2 \circ s^{-1}(\lambda_T)} = \frac{-2ps(d_T)}{(s')^2(d_T)} \sim \frac{-2pd_T}{(1+p)s'(d_T)}$$

since $m\iota\mu = -2\iota s''/(s')^2 \sim -2p/s'$ and $s \sim \iota s'/(1+p)$ by Karamata's theorem (see, e.g., Proposition 1.5.8 in Bingham et al. (1993)). It then follows from Lemma A1 (b) that

$$\frac{1}{\lambda_T(m_s\sigma_s^2)[\lambda_T]}\int_0^T X_t\mu(X_t)dt \to_d -\frac{1}{2}L(\tau, 0). \tag{A.6}$$

On the other hand, we can show that $m(\mu) = 0$ and $(m\mu)[\underline{x}, x]$ is square integrable as in the proof for $m$-I $\sigma^2$. Thus we have (A.4), and therefore, the stated result for $m$-BN $\sigma^2$ follows immediately from (A.6) and Lemma A2 (b).

□

*Proof for Lemma A3 (b).* We can show that $\iota\mu$ is $m$-SN if $\sigma^2$ is $m$-SN in similar ways in the proof for Lemma A3 (a). Then, by Lemma A1 (a), we have

$$\frac{1}{\lambda_T^2(m_s\iota_s\mu_s)(\lambda_T)}\int_0^T X_t\mu(X_t)dt \to_d \int_0^\tau \overline{m_s\iota_s\mu_s}(B_t)dt.$$

On the other hand, the $m$-integrabilities of $\iota$ and $\mu$ can be arbitrary. We will only prove both $\iota$ and $\mu$ are $m$-SN since other cases can be driven with a similar way. Let $\iota$ and $\mu$ be $m$-SN. Then the stochastic order of $\overline{X}_T\int_0^T \mu(X_t)dt$ is given by $\lambda_T^4(m_s^2\iota_s\mu_s)/T$ due to

Lemma A1 (a). However, we can deduce from (A.1) that

$$\frac{\lambda_T^4(m_s^2\iota_s\mu_s)/T}{\lambda_T^2(m_s\iota_s\mu_s)(\lambda_T)} = \frac{\lambda_T^2 m_s(\lambda_T)}{T} = \begin{cases} \lambda_T m_s(\lambda_T)/m_s[\lambda_T] \to 0, & \text{if } m \text{ is either I or BN} \\ 1, & \text{if } m \text{ is SN} \end{cases}$$

since $T = \lambda_T^2 m_s(\lambda_T)$ for SN $m$, and this provide the stated result. □

**Lemma A4.** *If DD holds, then*

$$\int_0^T (X_t - \overline{X}_T)\sigma(X_t)dW_t \prec_p \int_0^T (X_t - \overline{X}_T)\mu(X_t)dt$$

*for large $T$.*

*Proof for Lemma A4.* Note that if DD holds, then $\iota^2\sigma^2$ can be $m$-I, $m$-BN or $m$-SN. For each case, we can deduce from Lemma A1 that

$$\int_0^T (X_t - \overline{X}_T)\sigma(X_t)dW_t = \begin{cases} O_p(\sqrt{\lambda_T(m_s\iota_s^2\sigma_s^2)[\lambda_T]}), & \text{if } \iota^2\sigma^2 \text{ is } m\text{-I or } m\text{-BI} \\ O_p(\lambda_T(m_s^{1/2}\iota_s\sigma_s)(\lambda_T)), & \text{if } \iota^2\sigma^2 \text{ is } m\text{-SN} \end{cases} \tag{A.7}$$

where $(m_s\iota_s^2\sigma_s^2)[\lambda_T]$ is slowly varying when $\iota^2\sigma^2$ is $m$-I or $m$-BN. If $\iota^2\sigma^2$ is either $m$-I or $m$-BN, then we should have $m$-I $\sigma^2$. Then it follows from Lemma A3, (A.3) and (A.7) that $\sqrt{\lambda_T(m_s\iota_s^2\sigma_s^2)[\lambda_T]}/\lambda_T \to 0$, which provide the desired results for $m$-I or $m$-BN $\iota^2\sigma^2$.

Now let $\iota^2\sigma^2$ be $m$-SN. Then it follows from (A.2) that

$$\frac{\lambda_T(m_s^{1/2}\iota_s\sigma_s)(\lambda_T)}{\lambda_T(m_s\sigma_s^2)[\lambda_T]} = \frac{d_T}{s'(d_T)(1/s')[d_T]} \to 0 \tag{A.8}$$

since $m_s^{1/2}\iota_s\sigma_s = (\iota/s') \circ s^{-1}$ and $m_s\sigma_s^2[\lambda_T] = (1/s')[d_T]$. The stated result for $m$-SN $\iota^2\sigma^2$ follows from Lemma A3, (A.6), (A.7) and (A.8), which completes the proof.

□

**Lemma A5.** *If DD does not hold, then*

$$\int_0^T (X_t - \overline{X}_T)\sigma(X_t)dW_t \succsim_p \int_0^T (X_t - \overline{X}_T)\mu(X_t)dt$$

*for large $T$.*

*Proof for Lemma A5.* If DD does not hold, then $\iota^2\sigma^2$ is $m$-SN, and hence, the stochastic order of $\int_0^T (X_t - \overline{X}_T)\sigma(X_t)dW_t$ is given by $\lambda_T(m_s^{1/2}\iota_s\sigma_s)(\lambda_T)$ as shown in (A.7). Moreover,

$\iota\mu$ is also $m$-strongly nonintegrable, and the stochastic order of $\int_0^T (X_t - \overline{X}_T)\mu(X_t)dt$ is $\lambda_T^2(m_s\iota_s\mu_s)(\lambda_T)$ as shown in the proof of Lemma A3 (b). Then we have

$$\frac{\lambda_T^2(m_s\iota_s\mu_s)(\lambda_T)}{\lambda_T(m_s^{1/2}\iota_s\sigma_s)(\lambda_T)} = \frac{\lambda_T(m_s^{1/2}\mu_s)(\lambda_T)}{\sigma_s(\lambda_T)} = \frac{s(d_T)\mu(d_T)}{s'(d_T)\sigma^2(d_T)} \to \frac{-p}{2(1+p)}$$

since $-2d_T\mu(d_T)/\sigma^2(d_T) \to p$ by Assumption 3.1, and $s(d_T)/[d_T s'(d_T)] \to 1/(1+p)$ by Karamata's theorem, from which we have the stated result since $p \neq -1$.

□

**Lemma A6.** *If ST holds, then*

$$\int_0^T (X_t - \overline{X}_T)dX_t \sim_p -\frac{1}{2}\int_0^T \sigma^2(X_t)dt$$

*for large $T$.*

*Proof for Lemma A6.* We write

$$\int_0^T (X_t - \overline{X}_T)dX_t = \frac{1}{2}\left[(X_T^2 - X_0^2) - 2\overline{X}_T(X_T - X_0) - \int_0^T \sigma^2(X_t)dt\right]. \tag{A.9}$$

By applying Lemma A1 to the last term in (A.9), we have

$$\frac{1}{T_{\sigma^2}}\int_0^T \sigma^2(X_t)dt \to_d \begin{cases} L(\tau, 0), & \text{if DD holds} \\ \int_0^\tau \overline{m_s\sigma_s^2}(B_t)dt, & \text{otherwise.} \end{cases} \tag{A.10}$$

Moreover, we know that if ST holds, then

$$(X_T^2 - X_0^2) - 2\overline{X}_T(X_T - X_0) = o_p(d_T^2) \tag{A.11}$$

due to Lemma A2 (a). It then follows from (A.2) that $d_T^2 = o\left(\lambda_T(m_s\sigma_s^2)[\lambda_T]\right)$ when DD holds. On the other hand, if DD does not hold, then we have $d_T^2 = O(\lambda_T^2(m_s\sigma_s^2)(\lambda_T))$ because

$$\frac{d_T^2}{\lambda_T^2(m_s\sigma_s^2)(\lambda_T)} = \left[\frac{d_T s'(d_T)}{s(d_T)}\right]^2 \to (1+p)^2 \tag{A.12}$$

since $s(\lambda) \sim \lambda s'(\lambda)/(1+p)$ and $p \neq -1$. In all cases, we have $d_T^2 = O(T_{\sigma^2})$, and hence, the stated result follows from (A.9), (A.10) and (A.11). □

**Lemma A7.** *If DD holds, then*

$$\int_0^T (X_t - \overline{X}_T) dX_t \sim_p -\frac{1}{2}\int_0^T \sigma^2(X_t)dt$$

*for large $T$.*

*Proof for Lemma A7.* Due to Lemma A6, we only need to prove the statement for SN $m$. If $m$ is SN, then

$$\overline{X}_T = O_p(\lambda_T^2(m_s\iota_s)(\lambda_T)/T) = O_p(d_T)$$

by Lemma A1, and hence, we have

$$(X_T^2 - X_0^2) - 2\overline{X}_T(X_T - X_0) = O_p(d_T^2) \tag{A.13}$$

since $X_t = O_p(d_T)$ for all $t \in [0, T]$ by Kim and Park (2013). However, if DD holds, we have (A.2), and hence, the stated result follows from (A.9), (A.10) and (A.13). □

**Lemma A8.** *If $f$ is a twice differentiable function, then we have*

$$\sum_{i=1}^N f(X_{(i-1)\Delta})\Delta = \int_0^T f(X_t)dt + O_p(\Delta T(f'\mu)T) + O_p(\Delta T(f''\sigma^2)T) + O_p(\Delta T(f'\sigma)T^{1/2}))$$

*as $T \to \infty$ and $\Delta \to 0$.*

*Proof for Lemma A8.* We have

$$\begin{aligned}\sum_{i=1}^N f(X_{(i-1)\Delta})\Delta &= \int_0^T f(X_t)dt - \sum_{i=1}^N \int_{(i-1)\Delta}^{i\Delta} (f(X_t) - f(X_{(i-1)\Delta}))dt \\ &= \int_0^T f(X_t)dt - R_T^a - R_T^b \end{aligned} \tag{A.14}$$

by applying Itô's lemma to $f(X_t) - f(X_{(i-1)\Delta})$, where

$$\begin{aligned} R_T^a &= \sum_{i=1}^N \int_{(i-1)\Delta}^{i\Delta} \int_{(i-1)\Delta}^{t} \left( f'\mu(X_s) + \frac{f''\sigma^2(X_s)}{2} \right) dsdt, \\ R_T^b &= \sum_{i=1}^N \int_{(i-1)\Delta}^{i\Delta} \int_{(i-1)\Delta}^{t} f'\sigma(X_s)dW_s dt. \end{aligned}$$

For $R_T^a$, we have

$$
\begin{aligned}
|R_T^a| &= \left| \sum_{i=1}^{N} \int_{(i-1)\Delta}^{i\Delta} (i\Delta - s) \left( f'\mu(X_s) + \frac{f''\sigma^2(X_s)}{2} \right) ds \right| \\
&\leq \Delta \int_0^T \left| f'\mu(X_s) + \frac{f''\sigma^2(X_s)}{2} \right| ds \\
&= O_p(\Delta T(f'\mu)T) + O_p(\Delta T(f''\sigma^2)T) \qquad \text{(A.15)}
\end{aligned}
$$

by changing the order of the integrals.

As for $R_T^b$, we have

$$
R_T^b = \sum_{i=1}^{N} \int_{(i-1)\Delta}^{i\Delta} (i\Delta - s) f'\sigma(X_s) dW_s
$$

by changing the order of the integrals. Define a continuous martingale $D$ by

$$
D_t = \sum_{i=1}^{j-1} \int_{(i-1)\Delta}^{i\Delta} (i\Delta - s) f'\sigma(X_s) dW_s + \int_{(j-1)\Delta}^{t} (i\Delta - s) f'\sigma(X_s) dW_s
$$

for $t \in [(j-1)\Delta, j\Delta]$, $i = 1, 2, ...$, so that $D_T = R_T^b$. Then the quadratic variation $[D]$ of $D$ follows that

$$
[D]_T = \sum_{i=1}^{N} \int_{(i-1)\Delta}^{i\Delta} (i\Delta - s)^2 \left( f'\sigma(X_s) \right)^2 ds = O_p \left( \Delta^2 T \left( (f'\sigma)^2 \right) T \right),
$$

and therefore,

$$
R_T^b = O_p(\Delta T(f'\sigma)T^{1/2}). \qquad \text{(A.16)}
$$

The stated result follows from (A.14) with (A.15) and (A.16). □

**Lemma A9.** *Let $D$ be a continuous martingale defined as $D_t = \int_0^t f(X_s) dW_s$. If $\Delta T(f^2) \to 0$, then we have*

$$
\sup_{|t-s| \leq \Delta} |D_t - D_s| = O_p \left( \Delta^{1/2} T(f) \sqrt{\log 1/[\Delta T(f^2)]} \right)
$$

*as $T \to \infty$ and $\Delta \to 0$.*

*Proof for Lemma A9.* The quadratic variation of $D$ satisfies

$$\sup_{|t-s|\leq\Delta}[D]_t-[D]_s=\sup_{|t-s|\leq\Delta}\int_s^t f^2(X_r)dr=O_p(\Delta T(f^2)) \tag{A.17}$$

uniformly in $0\leq s\leq t\leq T$. Then, by the DDS (or Dambis, Dubins-Schwarz) theorem (see Revuz and Yor (1999), page 496), we may represent $D$ as $D_t=B\circ[D]_t$ where $B$ is the DDS Brownian motion of $D$. Thus we have

$$\begin{aligned}\sup_{|t-s|\leq\Delta}|D_t-D_s|&=\sup_{|t-s|\leq\Delta}|B\circ[D]_t-B\circ[D]_s|\\&=\sup_{|t-s|\leq\Delta}\left(2\left([D]_t-[D]_s\right)\log 1/([D]_t-[D]_s)\right)^{1/2}\end{aligned} \tag{A.18}$$

due to the modulus of continuity of Brownian motion (see Borodin and Salminen (1996), page 50). The stated result follows immediately from (A.17) and (A.18).

□

**Lemma A10.** *Let $f$ be a twice differentiable function. Then we have the following:*

*(a) We have*

$$\begin{aligned}\sum_{i=1}^N f(X_{(i-1)\Delta})\int_{(i-1)\Delta}^{i\Delta}g(X_t)dt=\int_0^T fg(X_t)dt+O_p(\Delta T(gf'\mu)T)+O_p(\Delta T(gf''\sigma^2)T)\\+O_p(\Delta T(gf'\sigma)T^{1/2}))\end{aligned}$$

*as $T\to\infty$ and $\Delta\to 0$.*

*(b) If $\Delta T\left((f'\sigma)^2\right)\to 0$, then we have*

$$\begin{aligned}&\sum_{i=1}^N f(X_{(i-1)\Delta})\int_{(i-1)\Delta}^{i\Delta}g(X_t)dW_t\\&=\int_0^T fg(X_t)dW_t+O_p\left(\Delta T\left(gf'\mu\right)T^{1/2}\right)+O_p\left(\Delta T\left(gf''\sigma^2\right)T^{1/2}\right)\\&\quad+O_p\left(\Delta^{1/2}T\left(gf'\sigma\right)T^{1/2}\sqrt{\log 1/[\Delta T\left((f'\sigma)^2\right)]}\right)\end{aligned}$$

*as $T\to\infty$ and $\Delta\to 0$.*

*Proof for Lemma A10.* For the first statement, we write

$$\begin{aligned}
&\left|\int_0^T fg(X_t)dt - \sum_{i=1}^N f(X_{(i-1)\Delta})\int_{(i-1)\Delta}^{i\Delta} g(X_t)dt\right| \\
&= \left|\sum_{i=1}^N \int_{(i-1)\Delta}^{i\Delta} g(X_t)\left[f(X_t) - f(X_{(i-1)\Delta})\right]dt\right| \\
&\leq T(g)\left|\sum_{i=1}^N \int_{(i-1)\Delta}^{i\Delta} \left[f(X_t) - f(X_{(i-1)\Delta})\right]dt\right|. \qquad \text{(A.19)}
\end{aligned}$$

However, it follows from Lemma A8 that

$$\left|\sum_{i=1}^N \int_{(i-1)\Delta}^{i\Delta} \left[f(X_t) - f(X_{(i-1)\Delta})\right]dt\right| = O_p(\Delta T(f'\mu)T) + O_p(\Delta T(f''\sigma^2)T) + O_p(\Delta T(f'\sigma)T^{1/2}))$$

which, in view of (A.19), completes the proof for the first part.

For the second statement, we have

$$\int_0^T fg(X_t)dW_t - \sum_{i=1}^N f(X_{(i-1)\Delta})\int_{(i-1)\Delta}^{i\Delta} g(X_t)dW_t = R_T^a + R_T^b \qquad \text{(A.20)}$$

by applying Itô's lemma to $f(X_t) - f(X_{(i-1)\Delta})$, where

$$\begin{aligned}
R_T^a &= \sum_{i=1}^N \int_{(i-1)\Delta}^{i\Delta} g(X_t)\int_{(i-1)\Delta}^t \left(f'\mu(X_s) + \frac{f''\sigma^2(X_s)}{2}\right)dsdV_t, \\
R_T^b &= \sum_{i=1}^N \int_{(i-1)\Delta}^{i\Delta} g(X_t)\int_{(i-1)\Delta}^t f'\sigma(X_s)dW_sdW_t.
\end{aligned}$$

Define two continuous martingale $D^a$ and $D^b$, respectively, by

$$\begin{aligned}
D_t^a =& \sum_{i=1}^{j-1} \int_{(i-1)\Delta}^{i\Delta} g(X_r)\int_{(i-1)\Delta}^r \left(f'\mu(X_s) + \frac{f''\sigma^2(X_s)}{2}\right)dsdV_r \\
&+ \int_{(j-1)\Delta}^t g(X_r)\int_{(j-1)\Delta}^r \left(f'\mu(X_s) + \frac{f''\sigma^2(X_s)}{2}\right)dsdV_r, \\
D_t^b =& \sum_{i=1}^{j-1} \int_{(i-1)\Delta}^{i\Delta} g(X_r)\int_{(i-1)\Delta}^r f'\sigma(X_s)dW_sdW_r + \int_{(j-1)\Delta}^t g(X_r)\int_{(j-1)\Delta}^r f'\sigma(X_s)dW_sdW_r
\end{aligned}$$

for $t \in [(j-1)\Delta, j\Delta]$, $i = 1, 2, ...$, so that $D_T^a = R_T^a$ and $D_T^b = R_T^b$. Then, for the quadratic

variations of $D^a$ and $D^b$, we have

$$\begin{aligned}[D^a]_T &= \sum_{i=1}^{N} \int_{(i-1)\Delta}^{i\Delta} g^2(X_t) \left[ \int_{(i-1)\Delta}^{t} \left( f'\mu(X_s) + \frac{f''\sigma^2(X_s)}{2} \right) ds \right]^2 dt \\ &= O_p\left(\Delta^2 T\left((gf'\mu)^2\right)T\right) + O_p\left(\Delta^2 T\left((gf''\sigma^2)^2\right)T\right) \end{aligned} \tag{A.21}$$

and

$$\begin{aligned}[D^b]_T &= \sum_{i=1}^{N} \int_{(i-1)\Delta}^{i\Delta} g^2(X_t) \left[ \int_{(i-1)\Delta}^{t} f'\sigma(X_s) dW_s \right]^2 dt \\ &= O_p\left(\Delta T\left((gf'\sigma)^2\right) T \log 1/[\Delta T\left((f'\sigma)^2\right)]\right) \end{aligned} \tag{A.22}$$

since

$$\sup_{|t-s|\le\Delta} \left| \int_s^t f'\sigma(X_r) dW_r \right| = O_p\left(\Delta^{1/2} T(f'\sigma)\sqrt{\log 1/[\Delta T\left((f'\sigma)^2\right)]}\right)$$

due to Lemma A9. Therefore, the state result for the second part follows immediately from (A.20), (A.21) and (A.22) with the constructions of $D^a$ and $D^b$. □

**Lemma A11.** *We have*

$$\begin{aligned} \sum_{i=1}^{N} \int_{(i-1)\Delta}^{i\Delta} \int_{(i-1)\Delta}^{t} f(X_s) ds g(X_t) dt &= O_p(\Delta T(fg)T), \\ \sum_{i=1}^{N} \int_{(i-1)\Delta}^{i\Delta} \int_{(i-1)\Delta}^{t} f(X_s) ds g(X_t) dW_t &= O_p(\Delta T(fg)T^{1/2}) \end{aligned}$$

*as $T \to \infty$ and $\Delta \to 0$.*

*Proof for Lemma A11.* We have

$$\left| \sum_{i=1}^{N} \int_{(i-1)\Delta}^{i\Delta} \int_{(i-1)\Delta}^{t} f(X_s) ds g(X_t) dt \right| \le T(fg) \sum_{i=1}^{N} \int_{(i-1)\Delta}^{i\Delta} \int_{(i-1)\Delta}^{t} ds dt = O_p(\Delta T(fg)T)$$

from which, we have the first statement.

For the second part, we define a continuous martingale $D$ by

$$D_t = \sum_{i=1}^{j-1} \int_{(i-1)\Delta}^{i\Delta} \int_{(i-1)\Delta}^{r} f(X_s) ds g(X_r) dW_r + \int_{(j-1)\Delta}^{t} \int_{(i-1)\Delta}^{r} f(X_s) ds g(X_r) dW_r$$

for $t \in [(j-1)\Delta, j\Delta]$, $i = 1, 2, ...$, so that

$$D_T = \sum_{i=1}^{N} \int_{(i-1)\Delta}^{i\Delta} \int_{(i-1)\Delta}^{t} f(X_s) ds g(X_t) dW_t.$$

Then, for the quadratic variation of $D$, we have

$$[D]_T = \sum_{i=1}^{N} \int_{(i-1)\Delta}^{i\Delta} \left( \int_{(i-1)\Delta}^{t} f(X_s) ds \right)^2 g^2(X_t) dt = O_p(\Delta^2 T((fg)^2)T)$$

which completes the proof of the second part. □

**Lemma A12.** *If $\Delta T(f^2) \to 0$, then we have*

$$\sum_{i=1}^{N} \int_{(i-1)\Delta}^{i\Delta} \int_{(i-1)\Delta}^{t} f(X_s) dW_s g(X_t) dt = O_p\left(\Delta^{1/2} T(fg) T \sqrt{1/\Delta T(f^2)}\right),$$

$$\sum_{i=1}^{N} \int_{(i-1)\Delta}^{i\Delta} \int_{(i-1)\Delta}^{t} f(X_s) dW_s g(X_t) dW_t = O_p\left(\Delta^{1/2} T(fg) T^{1/2} \sqrt{1/\Delta T(f^2)}\right)$$

*as $T \to \infty$ and $\Delta \to 0$.*

*Proof for Lemma A12.* The first part follows immediately from Lemma A9. For the second part, we define a continuous martingale $D$ by

$$D_t = \sum_{i=1}^{j-1} \int_{(i-1)\Delta}^{i\Delta} \int_{(i-1)\Delta}^{r} f(X_s) dW_s g(X_r) dW_r + \int_{(j-1)\Delta}^{t} \int_{(i-1)\Delta}^{r} f(X_s) dW_s g(X_r) dW_r$$

for $t \in [(j-1)\Delta, j\Delta]$, $i = 1, 2, ...$, so that

$$D_T = \sum_{i=1}^{N} \int_{(i-1)\Delta}^{i\Delta} \int_{(i-1)\Delta}^{t} f(X_s) dW_s g(X_t) dW_t.$$

Then, for the quadratic variation of $D$, we have

$$[D]_T = \sum_{i=1}^{N} \int_{(i-1)\Delta}^{i\Delta} \left( \int_{(i-1)\Delta}^{t} f(X_s) dW_s \right)^2 g^2(X_t) dt = O_p\left(\Delta T((fg)^2) T \log 1/[\Delta T(f^2)]\right)$$

due to Lemma A9, which completes the proof of the second part. □

**Lemma A13.** *We have*

$$\sum_{i=1}^{N}\left(\triangle x_i - \overline{\triangle x}_N\right)^2 = \sum_{i=1}^{N}\triangle x_i^2 + O_p(\Delta T(\mu^2)T) + O_p(\Delta T(\sigma^2)) + O_p(\Delta T(\mu\sigma)T^{1/2})$$

*as* $T \to \infty$ *and* $\Delta \to 0$*, where* $\overline{\triangle x}_N = N^{-1}\sum_{i=1}^{N}\triangle x_i$*. Moreover, if* $\Delta T(\sigma^2) \to 0$*, then*

$$\begin{aligned}\sum_{i=1}^{N}(\triangle x_i)^2 = &\int_0^T \sigma^2(X_t)dt + O_p(\Delta T(\mu^2)T) + O_p(\Delta T(\mu\sigma)T^{1/2}) \\ &+ O_p\left(\Delta^{1/2}T(\mu\sigma)T\sqrt{\log 1/[\Delta T(\sigma^2)]}\right) + O_p\left(\Delta^{1/2}T(\sigma^2)T^{1/2}\sqrt{\log 1/[\Delta T(\sigma^2)]}\right)\end{aligned}$$

*as* $T \to \infty$ *and* $\Delta \to 0$*.*

*Proof for Lemma A13.* We write

$$\sum_{i=1}^{N}\left(\triangle x_i - \overline{\triangle x}_N\right)^2 = \sum_{i=1}^{N}(\triangle x_i)^2 + N(\overline{\triangle x}_N)^2. \tag{A.23}$$

However, we have

$$\begin{aligned}\overline{\triangle x}_N &= \frac{1}{N}\left(X_T - X_0\right) = \frac{\Delta}{T}\left[\int_0^T \mu(X_t)dt + \int_0^T \sigma(X_t)dW_t\right] \\ &= \frac{\Delta}{T}\left[O_p(T(\mu)T) + O_p(T(\sigma)T^{1/2})\right],\end{aligned}$$

and hence,

$$N(\overline{\triangle x}_N)^2 = O_p(\Delta T(\mu^2)T) + O_p(\Delta T(\sigma^2)) + O_p(\Delta T(\mu\sigma)T^{1/2}) \tag{A.24}$$

which provides the first statement.

Now, for the first term in (A.23), we have

$$\begin{aligned}\sum_{i=1}^{N}\left(x_i - x_{i-1}\right)^2 - [X]_T &= \sum_{i=1}^{N}\left(\left(X_{i\Delta} - X_{(i-1)\Delta}\right)^2 - \left([X]_{i\Delta} - [X]_{(i-1)\Delta}\right)\right) \\ &= 2\sum_{i=1}^{N}\int_{(i-1)\Delta}^{i\Delta}\left(X_t - X_{(i-1)\Delta}\right)dY_t \\ &= 2\left[R_T^a + R_T^b + R_T^c + R_T^d\right]\end{aligned}$$

by Itô's formula, where

$$R_T^a = \sum_{i=1}^N \int_{(i-1)\Delta}^{i\Delta} \int_{(i-1)\Delta}^{t} \mu(X_s)ds\mu(X_t)dt = O_p(\Delta T(\mu^2)T),$$
$$R_T^b = \sum_{i=1}^N \int_{(i-1)\Delta}^{i\Delta} \int_{(i-1)\Delta}^{t} \mu(X_s)ds\sigma(X_t)dW_t = O_p(\Delta T(\mu\sigma)T^{1/2}),$$
$$R_T^a = \sum_{i=1}^N \int_{(i-1)\Delta}^{i\Delta} \int_{(i-1)\Delta}^{t} \sigma(X_s)dW_s\mu(X_t)dt = O_p\left(\Delta^{1/2}T(\mu\sigma)T\sqrt{\log 1/[\Delta T(\sigma^2)]}\right),$$
$$R_T^b = \sum_{i=1}^N \int_{(i-1)\Delta}^{i\Delta} \int_{(i-1)\Delta}^{t} \sigma(X_s)dW_s\sigma(X_t)dW_t = O_p\left(\Delta^{1/2}T(\sigma^2)T^{1/2}\sqrt{\log 1/[\Delta T(\sigma^2)]}\right)$$

due to Lemma A11 and Lemma A12, which completes the proof of the second part since $[X]_T = \int_0^T \sigma^2(X_t)dt$. □

# Appendix B Proofs for Theorems

*Proof for Lemma 4.1.* Due to Lemma A6 and A7, it suffice to prove that

$$\int_0^T (X_t - \overline{X}_T)dX_t \not\to_p -\frac{1}{2}\int_0^T \sigma^2(X_t)dt \tag{B.1}$$

when neither ST nor DD hold. To prove (B.1), we let $X$ be strongly nonstationary, and define $X^T$ by $X_t^T = X_{Tt}/d_T$. Then, it follows from Kim and Park (2013) that $X^T \to_d X^\circ$ in $C[0,1]$, where $X^\circ = \overline{s^{-1}}(B \circ A)$ with Brownian motion $B$ having local time $L$, and time change $A$ is defined as

$$A_t = \left\{ s \,\middle|\, \int \overline{m}(x)L(s,x)dx > t \right\}.$$

Then, by the continuous mapping theorem, we have

$$\frac{\overline{X}_T}{d_T} = \frac{1}{Td_T}\int_0^T X_t dt = \int_0^1 X_t^T dt \to_d \int_0^1 X_t^\circ dt \equiv \overline{X}_1^\circ \tag{B.2}$$

and hence

$$\frac{1}{d_T^2}\left[(X_T^2 - X_0^2) - 2\overline{X}_T(X_T - X_0)\right] \to_d \left[(X_1^{\circ 2} - X_0^{\circ 2}) - 2\overline{X}_1^\circ(X_1^\circ - X_0^\circ)\right]. \tag{B.3}$$

Moreover, if DD does not hold, then

$$\frac{1}{d_T^2}\int_0^T \sigma^2(X_t)dt \to_d \frac{1}{(1+p)^2}\int_0^\tau \overline{m_s\sigma_s^2}(B_t)dt \tag{B.4}$$

by Lemma A1 since $d_T^2 \sim (1+p)^2\lambda_T^2(m_s\sigma_s^2)(\lambda_T)$ as shown in (A.12). Thus we have

$$\begin{aligned}&\frac{1}{d_T^2}\int_0^T (X_t - \overline{X}_T)dX_t \\ &\to_d \frac{1}{2}\left[(X_1^{\circ 2} - X_0^{\circ 2}) - 2\overline{X}_1^{\circ}(X_1^{\circ} - X_0^{\circ}) - \frac{1}{(1+p)^2}\int_0^\tau \overline{m_s\sigma_s^2}(B_t)dt\right]\end{aligned} \tag{B.5}$$

from which we have (B.1), which complete the proof. □

*Proof for Lemma 4.2.* The statement in the lemma follows immediately from Lemma A4 and A5. □

*Proof for Theorem 4.3.* The stated results follows immediately from Lemma A1 (a), (b) and Lemma 4.1. □

*Proof for Theorem 4.4.* Note that if DD does not hold, then the limiting process $X^\circ$ of $X^T$ is semi-martingale with quadratic variation $[X^\circ]$ given by

$$[X^\circ]_t = \int_0^{A_t} \left((\overline{s^{-1}})'_-\right)^2 (B_s)ds,$$

where $(\overline{s^{-1}})'_-$ is the left-hand derivative of $\overline{s^{-1}}$. The reader is referred to Kim and Park (2013) for more detailed discussions on the limiting process $X^\circ$. If we show

$$\int_0^{A_t} \left((\overline{s^{-1}})'_-\right)^2 (B_s)ds = \frac{1}{(1+p)^2}\int_0^{A_t} \overline{m_s\sigma_s^2}(B_t)dt \tag{B.6}$$

for all $t \in [0,1]$, then the stated result follows from (B.5) since

$$\begin{aligned}&\int_0^1 (X_t^\circ - \overline{X}_T^\circ)dX_t^\circ \\ &= \frac{1}{2}\left[(X_1^{\circ 2} - X_0^{\circ 2}) - 2\overline{X}_1^{\circ}(X_1^{\circ} - X_0^{\circ}) - \frac{1}{(1+p)^2}\int_0^\tau \overline{m_s\sigma_s^2}(B_t)dt\right]\end{aligned}$$

by Itô's formula, since $A_1 = \tau$.

To show (B.6), we note that $m_s\sigma_s^2 = \left((s^{-1})'\right)^2$ since $m_s\sigma_s^2 = (1/s' \circ s^{-1})^2$ and $(s^{-1})' = 1/s' \circ s^{-1}$. Moreover, if $p \neq 0$, then $(s^{-1})'$ is ultimately monotone, and hence, $(s^{-1})'(x) \sim$

$s^{-1}(x)/[x(1+p)]$ due to monotone density theorem (see, e.g., Theorem 1.7.2 in Bingham et al. (1993)). On the other hand, if $p = 0$, then $(s^{-1})'$ is slowly varying. In both cases, we have $\overline{s^{-1\prime}}(x) = (1+p)(\overline{s^{-1}})'_-(x)$ for all $x \in \mathbb{R}\backslash\{0\}$ since

$$\begin{aligned}
\overline{(s^{-1})'}(x) &= a|x|^{-p/(1+p)}1\{x > 0\} + b|x|^{-p/(1+p)}1\{x \le 0\}, \\
(\overline{s^{-1}})'_-(x) &= \frac{1}{1+p}\left(a|x|^{-p/(1+p)}1\{x > 0\} + b|x|^{-p/(1+p)}1\{x \le 0\}\right)
\end{aligned}$$

for some constant $a, b \in \mathbb{R}$ such that $|a| + |b| = 1$. Therefore, we have (B.6), and this complete the proof.

□

*Proof for Lemma 5.1.* First, we consider $\hat{\beta}$. For the proof, we note that

$$\sum_{i=1}^{N} x_{i-1}\Delta = \int_0^T X_t dt + O_p(\Delta T(\mu)T) + O_p(\Delta T(\sigma)T^{1/2}), \tag{B.7}$$

$$\sum_{i=1}^{N} x_{i-1}^2\Delta = \int_0^T X_t dt + O_p(\Delta T(\iota\mu)T) + O_p(\Delta T(\sigma^2)T) + O_p(\Delta T(\iota\sigma)T^{1/2}) \tag{B.8}$$

due to Lemma A8. Moreover, we can deduce from Lemma A10 that

$$\begin{aligned}
\sum_{i=1}^{N} x_{i-1}\triangle x_i &= \sum_{i=1}^{N} X_{(i-1)\Delta}\int_{(i-1)\Delta}^{i\Delta}\mu(X_t)dt + \sum_{i=1}^{N} X_{(i-1)\Delta}\int_{(i-1)\Delta}^{i\Delta}\sigma(X_t)dW_t \\
&= \int_0^T X_t\mu(X_t)dt + \int_0^T X_t\sigma(X_t)dW_t + R_T^{\mu} + R_T^{\sigma} \\
&= \int_0^T X_t dX_t + R_T^{\mu} + R_T^{\sigma}
\end{aligned} \tag{B.9}$$

where

$$\begin{aligned}
R_T^{\mu} &= O_p(\Delta T(\mu^2)T) + O_p(\Delta T(\sigma^2)T^{1/2}), \\
R_T^{\sigma} &= O_p(\Delta T(\mu\sigma)T^{1/2}) + O_p\left(\Delta^{1/2}T(\sigma^2)T^{1/2}\sqrt{\log 1/[\Delta T(\sigma^2)]}\right).
\end{aligned}$$

Then the stated result for $\hat{\beta}$ follows immediately from (B.7), (B.8) and (B.9) with Assumption 5.1 and 5.2.

Now, we consider $t(\hat{\beta})$. We write

$$\sum_{i=1}^{N}\left[\triangle x_i-(\hat{\alpha}-\hat{\beta}x_{i-1})\Delta\right]^2=\sum_{i=1}^{N}\left(\triangle x_i-\overline{\triangle x}_N\right)^2-\frac{\left[\sum_{i=1}^{N}\left(x_{i-1}-\overline{x}_N\right)\triangle x_i\right]^2}{\sum_{i=1}^{N}\left(x_{i-1}-\overline{x}_N\right)^2}, \tag{B.10}$$

where $\overline{x}_N=N^{-1}\sum_{i=1}^{N}x_{i-1}$.

For the second term in (B.10), we can deduce from (B.7), (B.8) and (B.9) with Assumption 5.1 and 5.2 that

$$\frac{\left[\sum_{i=1}^{N}\left(x_{i-1}-\overline{x}_N\right)\triangle x_i\right]^2}{\sum_{i=1}^{N}\left(x_{i-1}-\overline{x}_N\right)^2}=\Delta\frac{\left[\int_0^T(X_t-\overline{X}_T)dX_t\right]^2}{\int_0^T(X_t-\overline{X}_T)^2dt}(1+o_p(1)). \tag{B.11}$$

Moreover, we have

$$\begin{aligned}\Delta^{1/2}\int_0^T(X_t-\overline{X}_T)dX_t&=\Delta^{1/2}\int_0^T(X_t-\overline{X}_T)\mu(X_t)dt+\Delta^{1/2}\int_0^T(X_t-\overline{X}_T)\sigma(X_t)dW_t\\&=O_p(\Delta^{1/2}T(\iota\mu)T)+O_p(\Delta^{1/2}T(\iota\sigma)T^{1/2})=o_p(1)\end{aligned} \tag{B.12}$$

due, in particular, to Assumption 5.2. It follows from (B.10), (B.11) and (B.12) with Lemma A13 that

$$\frac{\hat{\sigma}^2}{\Delta}=\frac{1}{T}\sum_{i=1}^{N}\left[\triangle x_i-(\hat{\alpha}-\hat{\beta}x_{i-1})\Delta\right]^2=\frac{1}{T}\int_0^T\sigma^2(X_t)dt(1+o_p(1)) \tag{B.13}$$

due to Assumption 5.1 and 5.2. The stated result for $t(\hat{\beta})$ follows immediately from (B.13) with the proof of the first part of this lemma.

□

*Proof for Theorem 5.2.* For a strongly nonstationary $X$, we have

$$\frac{1}{Td_T^2}\int_0^T(X_t-\overline{X}_T)^2dt\to_d\int_0^1(X_t^\circ-\overline{X}_1^\circ)^2dt \tag{B.14}$$

in a similar way of (B.2). Note in particular that $Td_T^2=\lambda_T^2(m_s\iota_s^2)(\lambda_T)$. On the other hand, if $X$ is stationary or barely nonstationary, then we have

$$\frac{1}{T_{\iota^2}}\int_0^T(X_t-\overline{X}_T)^2dt\begin{cases}\to_p 1-[\pi(\iota)]^2/\pi(\iota^2), & \text{if } \iota^2 \text{ is } m\text{-I}\\ \to_p 1, & \text{if } \iota^2 \text{ is } m\text{-BI}\\ \to_d \int_0^\tau\overline{m_s\sigma_s^2}(B_t)dt, & \text{if } \iota^2 \text{ is } m\text{-SN}\end{cases} \tag{B.15}$$

by Lemma A1. In the following, we first prove the case where (a) neither ST nor DD holds, and then we consider the case where (b) either ST or DD holds.

**Part (a)** We let nether ST nor DD hold. Then it follows immediately from Lemma 5.1, Theorem 4.4 and (B.14) that

$$T\Delta^{-1}\hat{\beta} \sim_p \frac{\frac{1}{d_T^2}\int_0^T (X_t - \overline{X}_T)dX_t}{\frac{1}{Td_T^2}\int_0^T (X_t - \overline{X}_T)^2 dt} \to_d \frac{\int_0^1 (X_t^\circ - \overline{X}_1^\circ)dX_t^\circ}{\int_0^1 (X_t^\circ - \overline{X}_1^\circ)^2 dt}$$

as $T \to \infty$ and $\Delta \to 0$. Moreover, it follows from Lemma 5.1, Theorem 4.4, (B.14) and (B.4) with (B.6) that

$$t(\hat{\beta}) \sim_p \frac{\frac{1}{d_T^2}\int_0^T (X_t - \overline{X}_T)dX_t}{\left(\frac{1}{d_T^2}\int_0^T \sigma^2(X_t)dt\right)^{1/2}\left(\frac{1}{Td_T^2}\int_0^T (X_t - \overline{X}_T)^2 dt\right)^{1/2}} \to_d \frac{\int_0^1 (X_t^\circ - \overline{X}_1^\circ)dX_t^\circ}{[X^\circ]_1^{1/2}\left(\int_0^1 (X_t^\circ - \overline{X}_1^\circ)^2 dt\right)^{1/2}}$$

as $T \to \infty$ and $\Delta \to 0$, which completes the proof for the first case where nether ST nor DT holds.

**Part (b)** Now let either ST or DD holds. Then we may write

$$T\Delta^{-1}\hat{\beta} \sim_p \frac{TT_{\sigma^2}}{T_{\iota^2}} \frac{\frac{1}{T_{\sigma^2}}\int_0^T (X_t - \overline{X}_T)dX_t}{\frac{1}{T_{\iota^2}}\int_0^T (X_t - \overline{X}_T)^2 dt} \tag{B.16}$$

and

$$t(\hat{\beta}) \sim_p \left(\frac{TT_{\sigma^2}}{T_{\iota^2}}\right)^{1/2} \frac{\frac{1}{T_{\sigma^2}}\int_0^T (X_t - \overline{X}_T)dX_t}{\left(\frac{1}{T_{\sigma^2}}\int_0^T \sigma^2(X_t)dt\right)^{1/2}\left(\frac{1}{T_{\iota^2}}\int_0^T (X_t - \overline{X}_T)^2 dt\right)^{1/2}}. \tag{B.17}$$

If we show $TT_{\sigma^2}/T_{\iota^2} \to \infty$, then the stated result follows immediately from Lemma 5.1, Theorem 4.3, (B.14), (B.15) and (A.10) with (B.16) and (B.17).

To complete the proof, we note that

$$\frac{TT_{\sigma^2}}{T_{\iota^2}} = \begin{cases} T(m_s\sigma_s^2)[\lambda_T]/(m_s\iota_s^2)[\lambda_T], & \text{if } \sigma^2 \text{ is } m\text{-I or } m\text{-BN and } \iota^2 \text{ is } m\text{-I or } m\text{-BN}, \\ T\lambda_T(m_s\sigma_s^2)(\lambda_T)/(m_s\iota_s^2)[\lambda_T], & \text{if } \sigma^2 \text{ is } m\text{-SN and } \iota^2 \text{ is } m\text{-I or } m\text{-BN}, \\ T(m_s\sigma_s^2)[\lambda_T]/\lambda_T(m_s\iota_s^2)(\lambda_T), & \text{if } \sigma^2 \text{ is } m\text{-I or } m\text{-BN and } \iota^2 \text{ is } m\text{-SN}, \\ T(m_s\sigma_s^2)(\lambda_T)/(m_s\iota_s^2)(\lambda_T), & \text{if both } \sigma^2 \text{ and } \iota^2 \text{ are } m\text{-SN}. \end{cases}$$

Recall that $(m_s f_s)[\lambda]$ is slowly varying if $f$ is either $m$-I or $m$-BN. Thus if $\iota^2$ is either $m$-I or $m$-BN, then $TT_{\sigma^2}/T_{\iota^2} \to \infty$. Thus we only need to examine the cases where $\iota^2$ is $m$-SN.

If $\sigma^2$ is either $m$-I or $m$-BN, then

$$\frac{T(m_s\sigma_s^2)[\lambda_T]}{\lambda_T(m_s\iota_s^2)(\lambda_T)} = \left(\frac{T}{\lambda_T d_T m(d_T)}\right)\left(\frac{s'(d_T)(m_s\sigma_s^2)[\lambda_T]}{d_T}\right) \equiv a_T b_T$$

since $m_s\iota_s^2 = (\iota^2 m/s') \circ s^{-1}$. Moreover, we have $b_T \to \infty$ due to (A.2) and

$$a_T = \begin{cases} m_s[\lambda_T]/d_T m(d_T) \to \infty, & \text{if ST holds} \\ \lambda_T m_s(\lambda_T)/d_T m(d_T) \to 1, & \text{otherwise} \end{cases}$$

by Karamata's theorem.

Now let $\sigma^2$ be $m$-SN. Then ST should be satisfied, and we have

$$\begin{aligned} \frac{T(m_s\sigma_s^2)(\lambda_T)}{(m_s\iota_s^2)(\lambda_T)} &= \frac{\lambda_T m_s[\lambda_T]\sigma_s^2(\lambda_T)}{d_T^2} = \frac{s(d_T)m[d_T]\sigma^2(d_T)}{d_T^2} \\ &\sim \frac{s'(d_T)m[d_T]\sigma^2(d_T)}{(1+p)d_T} = \frac{m[d_T]}{(1+p)d_T m(d_T)} \to \infty \end{aligned}$$

by Karamata's theorem. Therefore, we have $TT_{\sigma^2}/T_{\iota^2} \to \infty$ in all cases, from which we have the stated result.

□

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